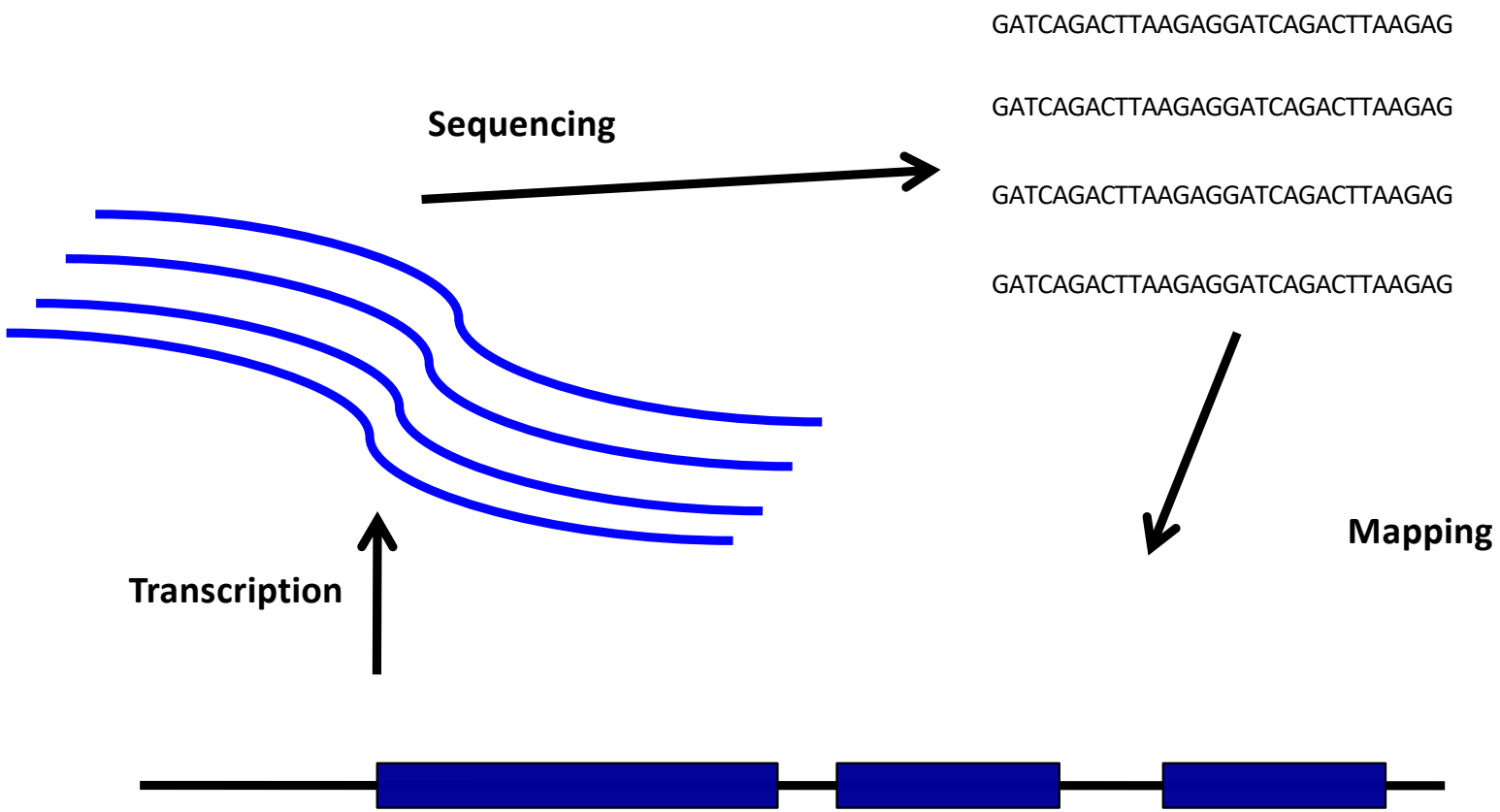
RNA-seq Quantification

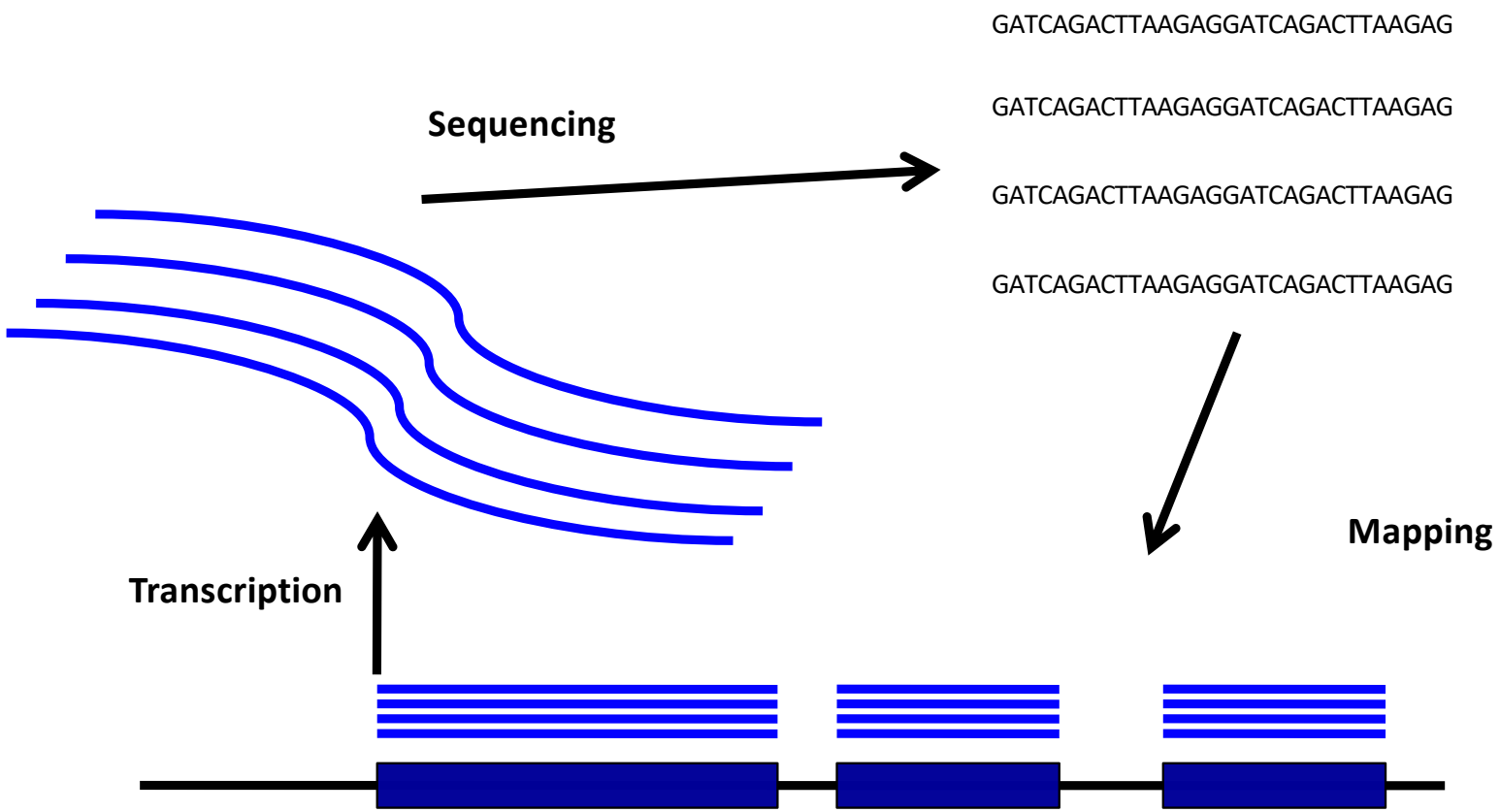
Outline

- short-read RNA-seq
- quantification:
  - read counts per gene
  - generative models: EM, effective length, TPM
  - uncertainty of expression values, and impact of sequencing depth
  - pseudo-alignment and equivalence classes
- biases and special protocols (repeats, 3’-tag, long reads)
- in practice: annotation, pipeline, QC
- the RNA-seq setting: normalizing fractions

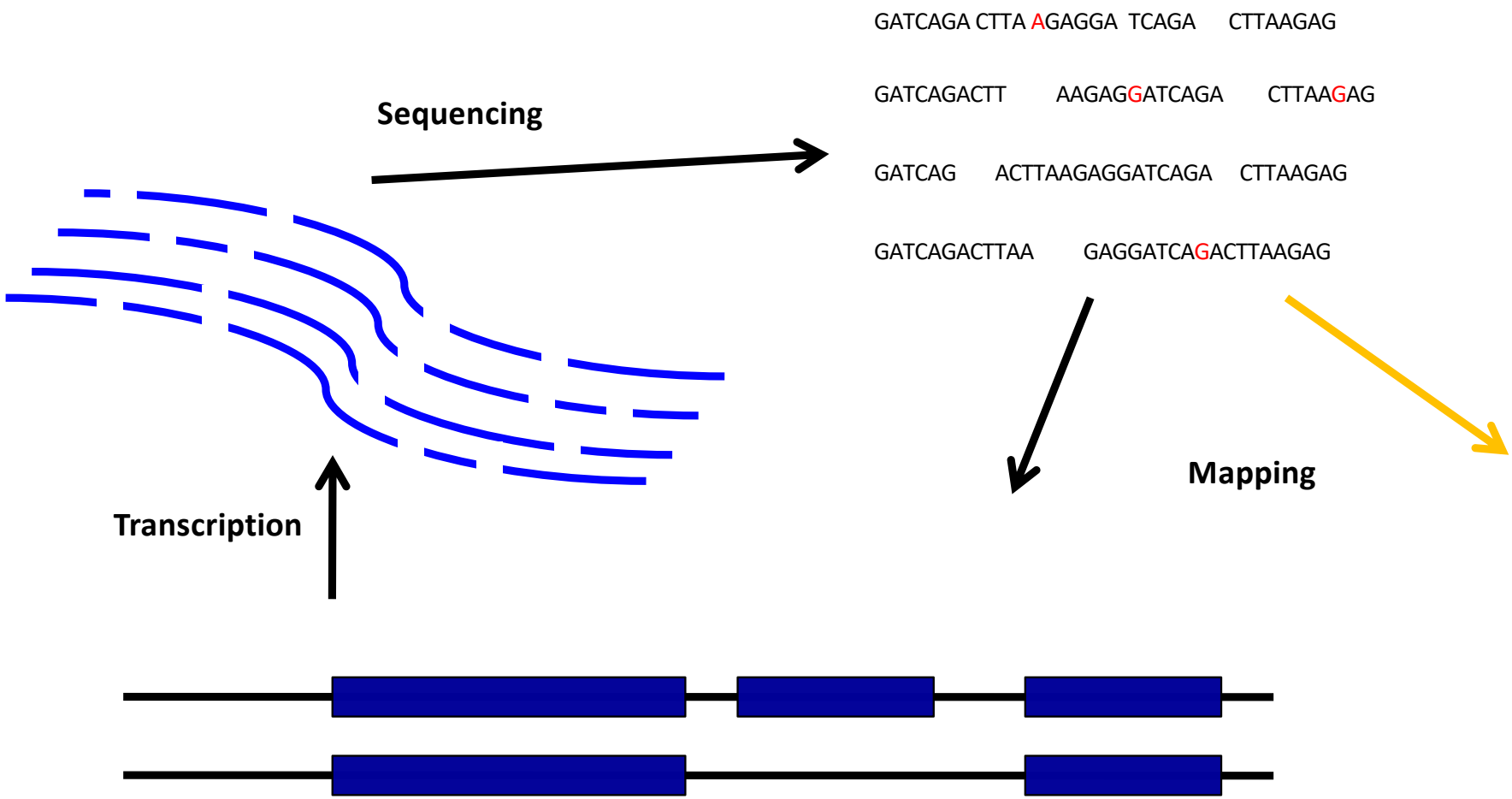
Transcript Expression with NGS



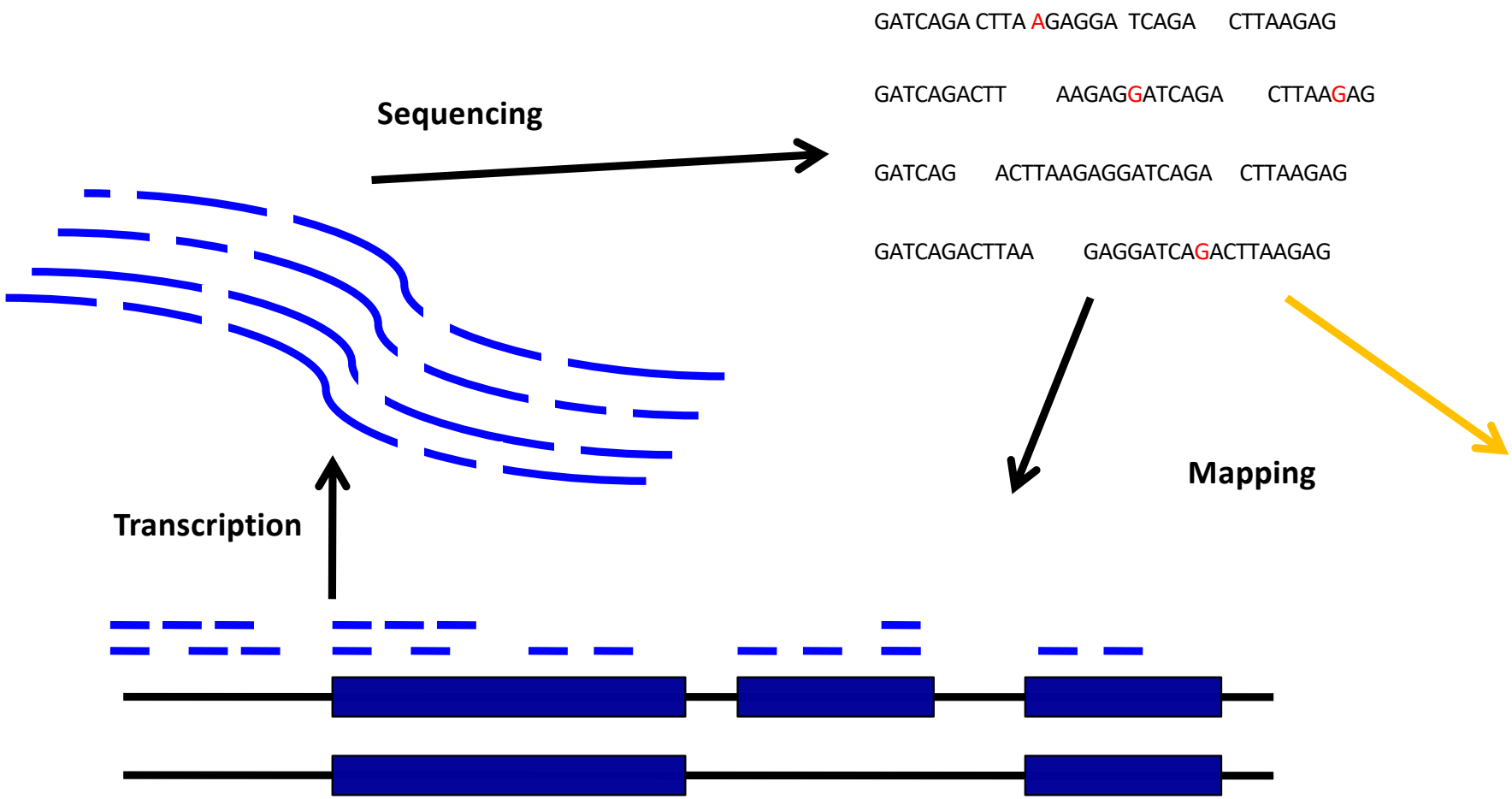
Read mapping and counting (ideal)



Read mapping and counting (short reads with SNPs and sequencing errors)



Read mapping and counting (short reads with SNPs and sequencing errors)



Abundance estimates

Abundance of what?

- Biologically relevant:
  - **gene level:**
    - # molecules transcribed from one gene locus (per cell)
  - **isoform level:**
    - # molecules of a specific isoform transcribed from one gene (per cell)
- Feasible with RNA-seq:
  - **relative fractions that indicate the abundance relative to all other genes/isoforms**

Gene-level Read Counts

Straightforward to compute:

- **Reads that map uniquely** to a gene locus
  → scales with gene length (cancels when the gene is compared across samples)
  → discards multi-mappers, so gene families are under-counted
- **All reads** that map to the gene locus, including multi-mappers
  → splitting multi-mappers properly needs abundance estimates of the genes; if abundance estimates are not available, heuristics are used:
  - assign each multi-mapper at random to one of the matching genes
  - fractional assignment: 1/#genes to each matching gene

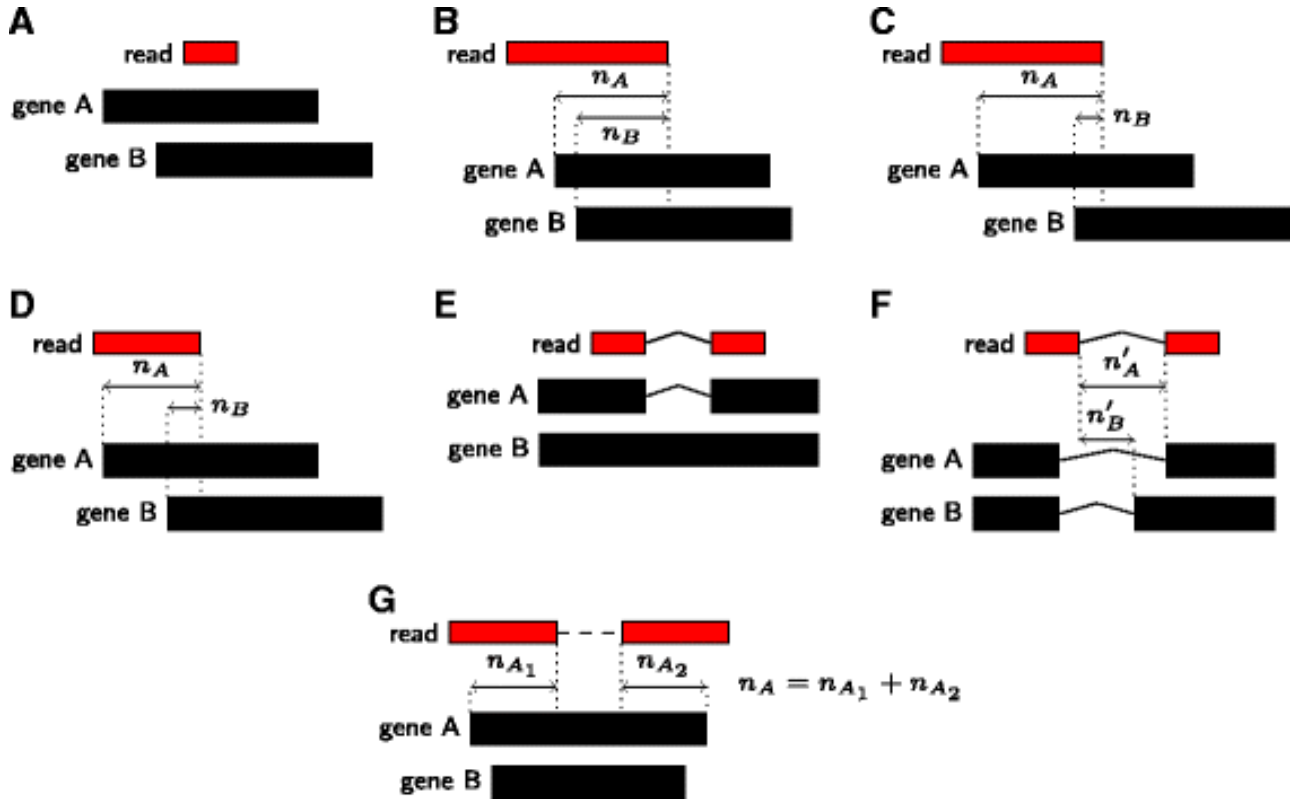
Rsubread::featureCounts – assigning reads to genes

- versatile function to count reads towards exons, transcripts, genes, …
- implements many different counting modes
- covers different aspects of overlap situations
  - partial overlap
  - overlapping multiple features at the same alignment position
  - overlapping multiple features at different alignment positions

- Default rule: a read counts for a gene if ≥ 1 aligned base overlaps one of its exons; whether its splice junctions match the annotation is not checked

Model-free Counting of Overlapping reads – Count Modes

mmquant: reads that map to several genes are counted as a merged feature (geneA--geneB)



Zytnicki 2017, BMC Bioinformatics (mmquant)

Counting Reads per Gene: Check the Defaults

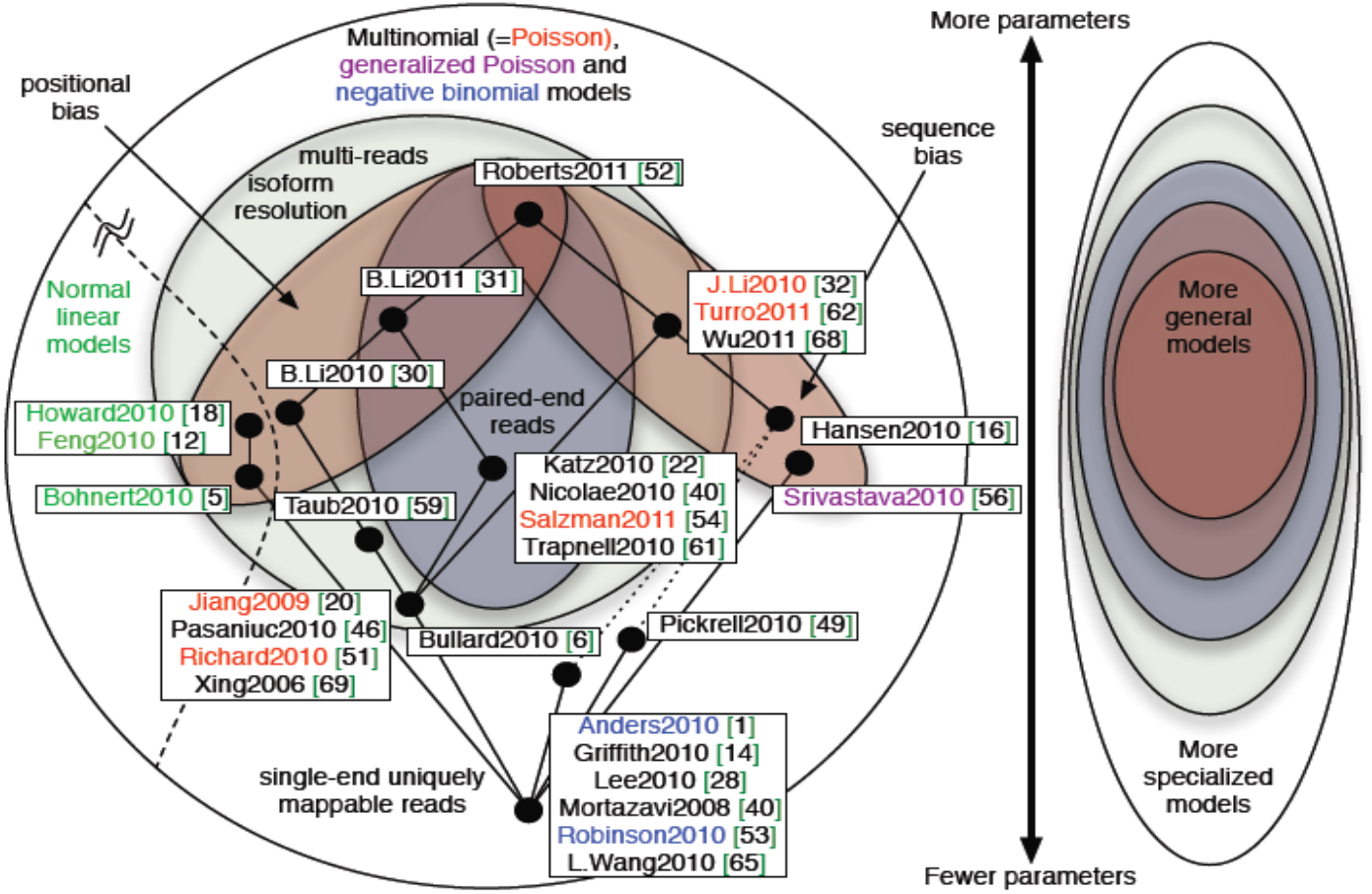
Setting	If you leave the default	Rsubread::featureCounts (R)	featureCounts (command line)
Strandedness	unstranded counting of a stranded library: antisense reads are counted	`strandSpecific=0`	`-s 0`
Paired-end reads	each mate is counted: one fragment gives two counts	`isPairedEnd=FALSE`	counts reads; for fragments: `-p --countReadPairs`
Multi-mapping reads	R: one count per alignment; command line: not counted	`countMultiMappingReads=TRUE, fraction=FALSE`	not counted; to count: `-M` or `-M --fraction`
Read overlapping two genes	not counted (ambiguous)	`allowMultiOverlap=FALSE`	not counted; for both genes: `-O`
Minimum overlap	1 base is enough; splice junctions are not checked	`minOverlap=1`	`--minOverlap 1`

→ counting multi-mappers with `fraction=FALSE` means the total counts from N reads can be larger than N

→ minOverlap=1 is not really stringent

Rsubread reference manual; Subread users guide (featureCounts)

Generative Models for RNA-seq quantification



Pachter 2011, https://doi.org/10.48550/arXiv.1104.3889

RNA-seq model

Probability that a read/fragment comes from transcript *t*:

$$\alpha_t = \mathrm{P}[\text{read from transcript } t] = \frac{1}{Z}\rho_t l_t$$

with:

ρ_t expression level / abundance / fraction

l_t transcript length

$Z = \sum_t \rho_t l_t$ normalization factor

Z = abundance-weighted mean transcript length. Later, l_t is replaced by the effective length.

RNA-seq model

Estimated probability that a read comes from transcript t:

$$\hat{\alpha}_t = \frac{X_t}{N} = \frac{\#\text{reads mapping to transcript } t}{\#\text{mappable reads in total}}$$

Abundance estimates:

$$\hat{\rho}_t \propto \frac{\hat{\alpha}_t}{l_t}$$

Maximum Likelihood Estimation

- If every read is assigned to exactly one transcript, the estimate on the previous slide is the unique maximum-likelihood estimate (MLE)

$$\text{with } \alpha = \{\alpha_t\}_{t \in T}$$

$$L[\alpha] = \prod_{t \in T} \prod_{f \in F_t} P[f \in t] \frac{1}{l_t}$$

$$= \prod_{t \in T} \prod_{f \in F_t} \alpha_t \frac{1}{l_t}$$

$$= \prod_{t \in T} \left(\frac{\alpha_t}{l_t} \right)^{X_t}$$

Effective Transcript Length

- Fragments have a length, so a transcript offers fewer start positions than its length: the read probabilities depend on an *effective* length:

$$l_t := \text{transcript length - fragment length + 1}$$

- With a fragment-length distribution D: effective length = transcript length – μ_t + 1, where μ_t is the mean of D truncated at the transcript length (otherwise short transcripts would get a length ≤ 0)
- We keep writing l_t, but always mean the effective length
- It is the part of the transcript that can yield fragments mapping back to it
  - so it should also reflect mappability, which depends on the aligner, sequence variants, …

Multi-reads

- Reads that cannot be uniquely assigned to one transcript were ignored so far
- Multi-reads can occur
  - if a read aligns more than once in the genome
  - if at an alignment position there is more than one transcript defined
- Multi-reads are not chance events: they come from exons shared by isoforms and from homologous sequence (gene families, pseudogenes, repeats)

Considering Multi-reads

- Define a compatibility matrix

$$\mathbf{Y} = \{y_{ft}\}_{f \in F, t \in T}$$

with

$$y_{ft} = \begin{cases} 1 \text{ if read } f \text{ aligns to transcript } t \\ 0 \text{ else} \end{cases}$$

- The likelihood is now:

$$\mathrm{L}[\alpha] = \prod_f \left(\sum_t y_{ft} \frac{\alpha_t}{l_t} \right)$$

- but now abundances must be estimated iteratively

Iterative Estimation

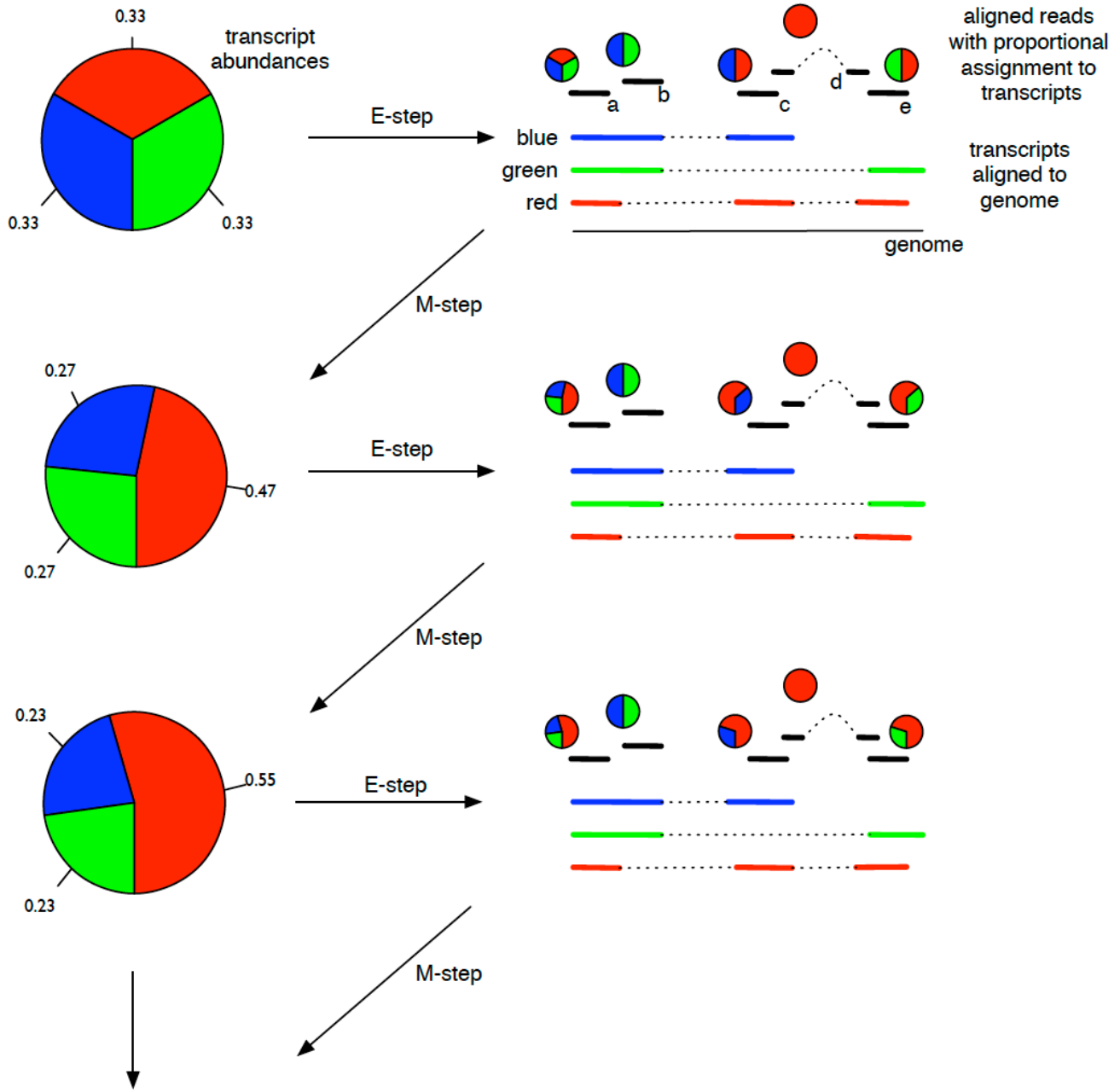
Expectation–maximization (EM):

1. Start: give every transcript a positive abundance (e.g. all equal) — *not* the unique-read estimate: a transcript without unique reads would stay at 0
2. E-step: split each multi-read between its compatible transcripts in proportion to α_t / l_t
3. M-step: recompute α_t from the (fractional) read counts
4. Repeat 2–3 until α stops changing

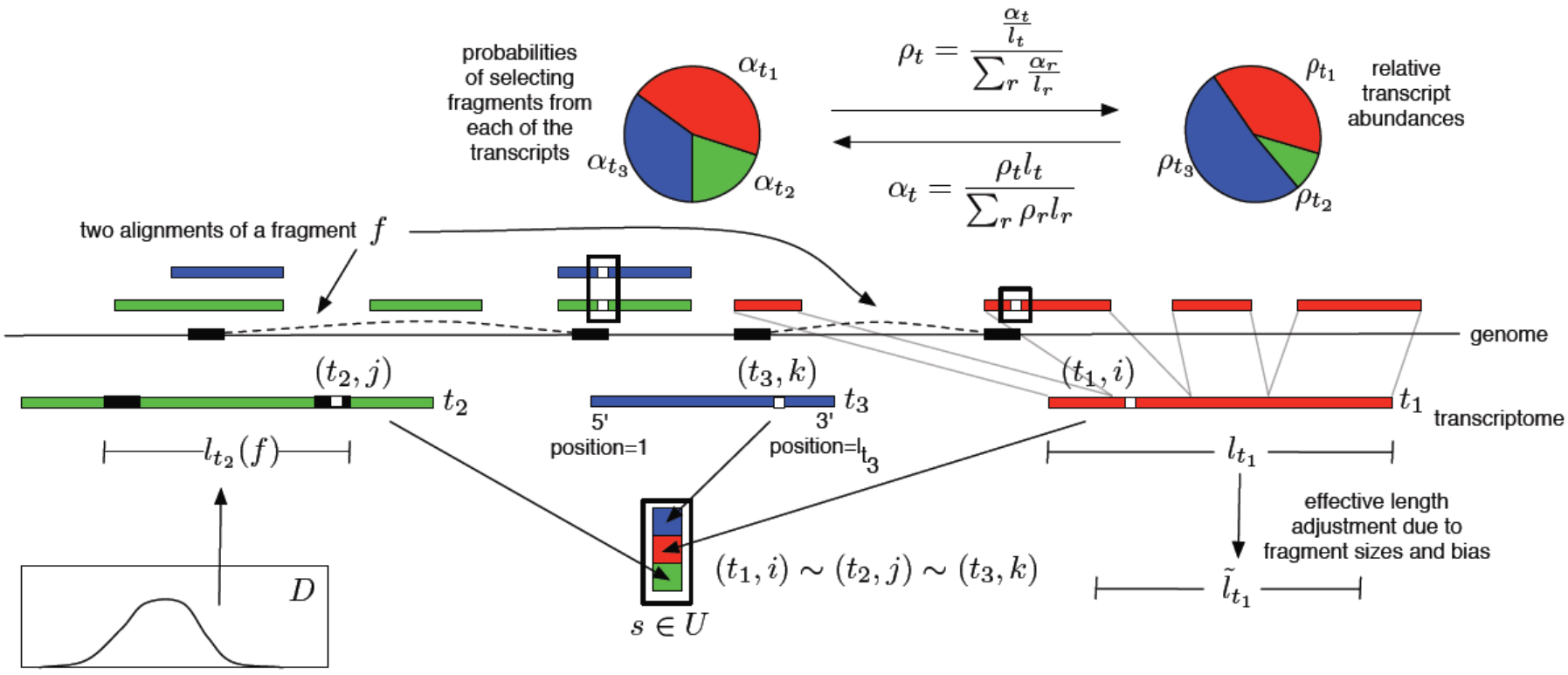
The log-likelihood is concave: EM reaches the global maximum. The maximum is not unique if no read distinguishes two transcripts.

ETH zürich University of Zurich[UZH]

Expectation-Maximization Estimation



Read Fractions α and Transcript Abundances ρ



Definition of expression levels

- Goal: Start from read counts and define a quantity that indicates **relative molar concentration of a transcript**

- Transcripts Per Million Transcripts

$$\text{TPM for transcript } t = 10^6 \times Z \times \frac{X_t}{l_t N}$$

- TPM is unitless and is a measure of relative abundance

- TPM assumes that the total sum of all counts (N) can be used to normalize across samples

RPKM / FPKM vs. TPM

- **RPKM** (reads per kilobase per million reads): $RPKM_t = 10^9 \cdot X_t / (l_t \cdot N)$; FPKM: the same with fragments
- **TPM**: $TPM_t = 10^6 \cdot (X_t / l_t) / \Sigma_r (X_r / l_r) = 10^6 \cdot RPKM_t / \Sigma_r RPKM_r$
  - Within one sample both rank transcripts identically: they differ by one factor per sample
- TPM sums to 10^6 in every sample; the sum of RPKM varies, so RPKM is not a relative concentration (Wagner, Kin & Lynch 2012)

→**Prefer TPM: in every sample it is a share of the same total**

- **Both divide by a total: if the composition differs (tissues, protocols), equal TPM does not mean equal concentration (later in this lecture)**

Zhao, Ye & Stanton 2020, RNA 26:903–909

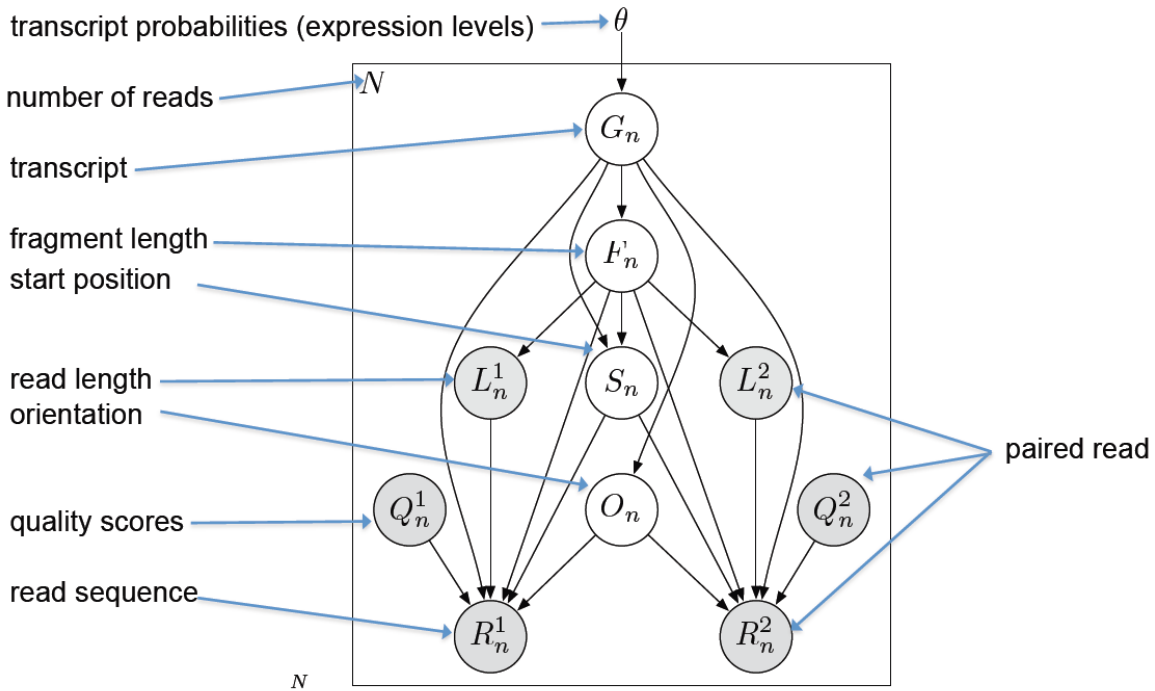
General Formulation of Abundance Estimation

A full model for the abundance estimation would ideally consider:

- position bias
- fragment-length distribution
- sequencing errors
- sequence-specific (random-hexamer priming) and GC bias
- …

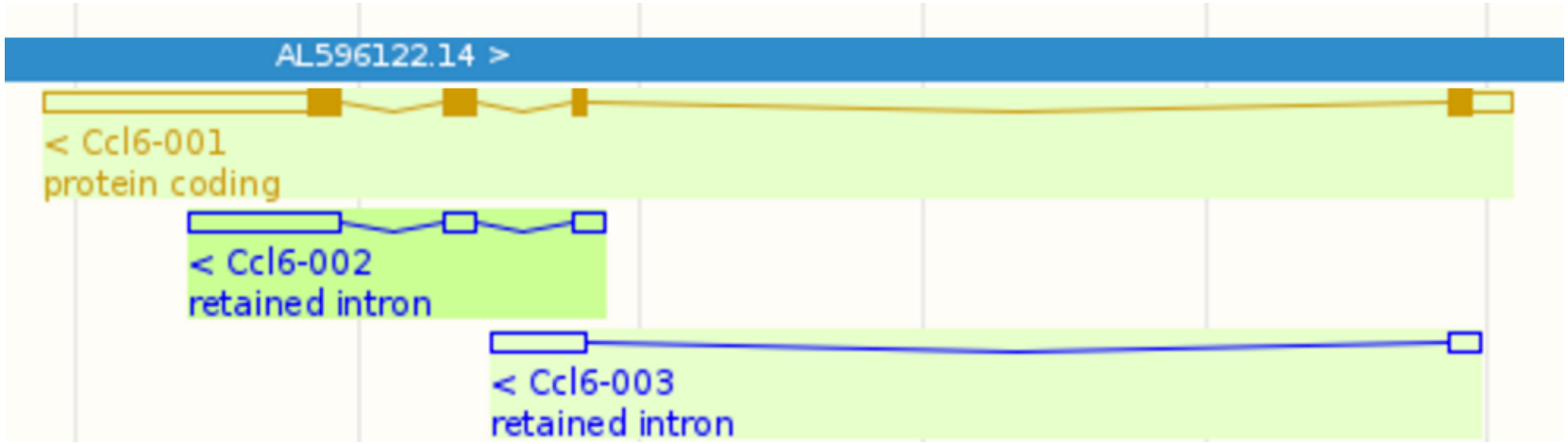
RSEM: Model specification

Li & Dewey 2011, BMC Bioinformatics



$$P(\mathbf{g},\mathbf{f},\mathbf{s},\mathbf{o},\ell,\mathbf{q},\mathbf{r}|\theta)=\prod_{n=1}^{N}P(g_n|\theta)P(f_n|g_n)P(s_n|f_n,g_n)P(o_n|g_n)P(q_n)P(\ell_n|f_n)P(r_n|g_n,f_n,s_n,o_n,\ell_n,q_n)$$

Example: RSEM

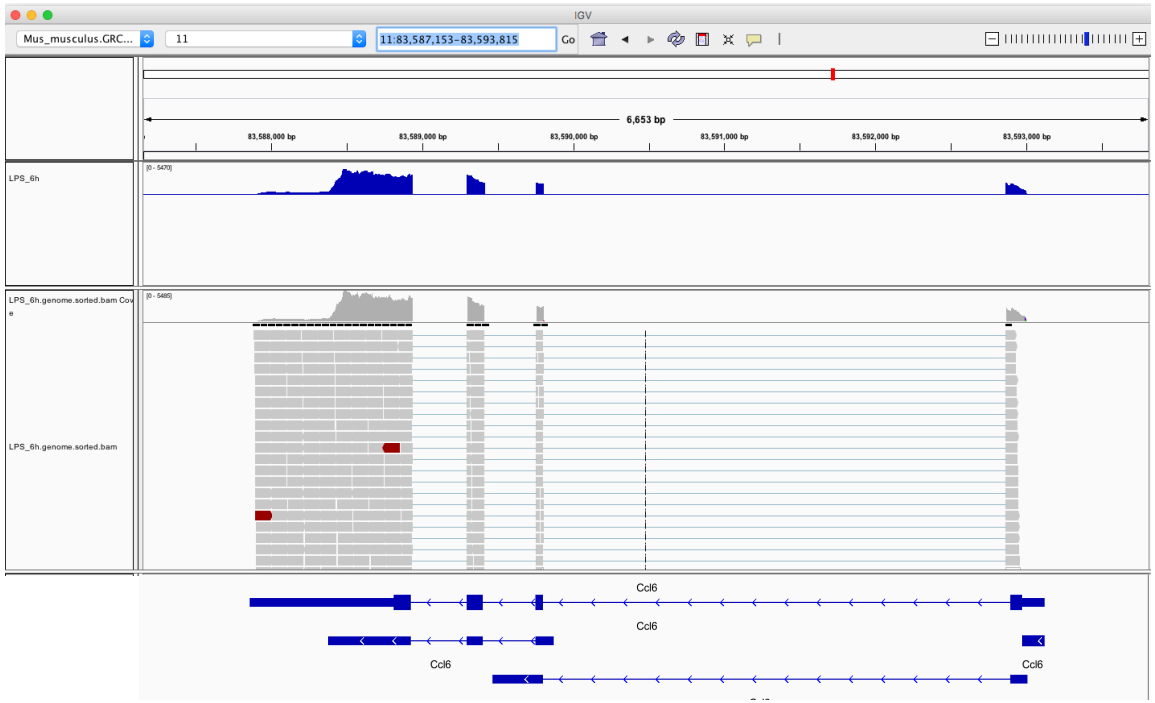


Ccl6 gene locus with 3 isoforms

follows the example:

https://github.com/bli25broad/RSEM_tutorial

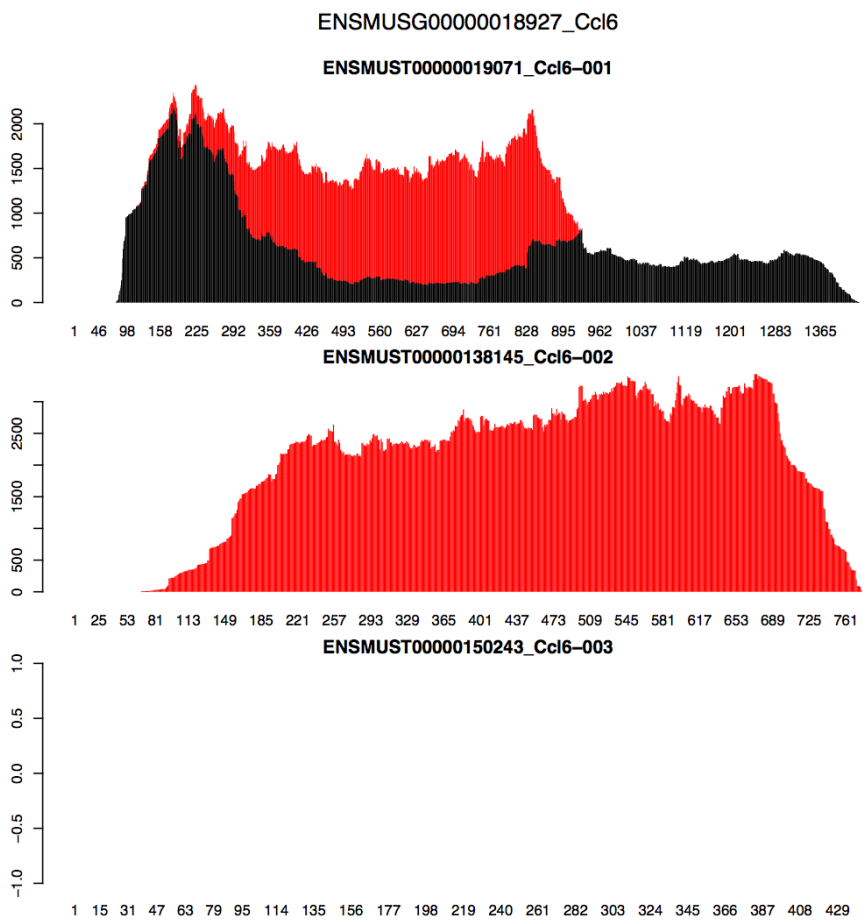
RSEM Result for Ccl6



transcript	length	eff. length	expected count	TPM	FPKM	IsoPct
Ccl6-001	1440	1194.85	7805.95	8862.10	9334.46	31
Ccl6-002	776	530.94	7719.05	19721.39	20772.55	69
Ccl6-003	442	202.64	0	0	0	0

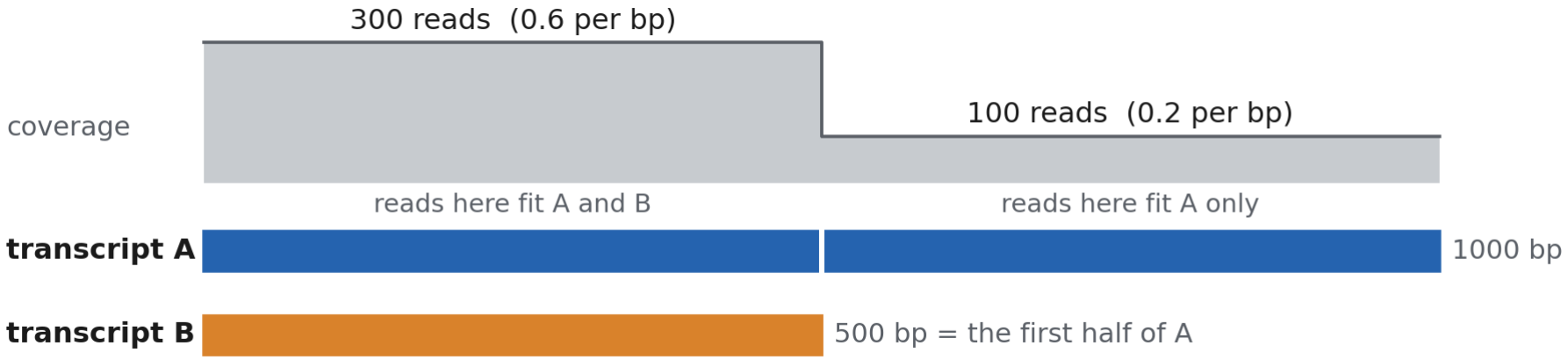
Expected counts are almost equal, yet the TPM of Ccl6-002 is 2.2 × higher. Why?

Ccl6 coverage in transcript space



- Ccl6 is on the minus strand: these plots run 5’ → 3’ along the transcript, mirrored relative to the genome view
- black: depth from unique alignments
- red: expected depth from multi-mapping reads, as assigned by RSEM
- **Ccl6-002 has no black at all: its 69 % rests entirely on how the model splits shared reads**

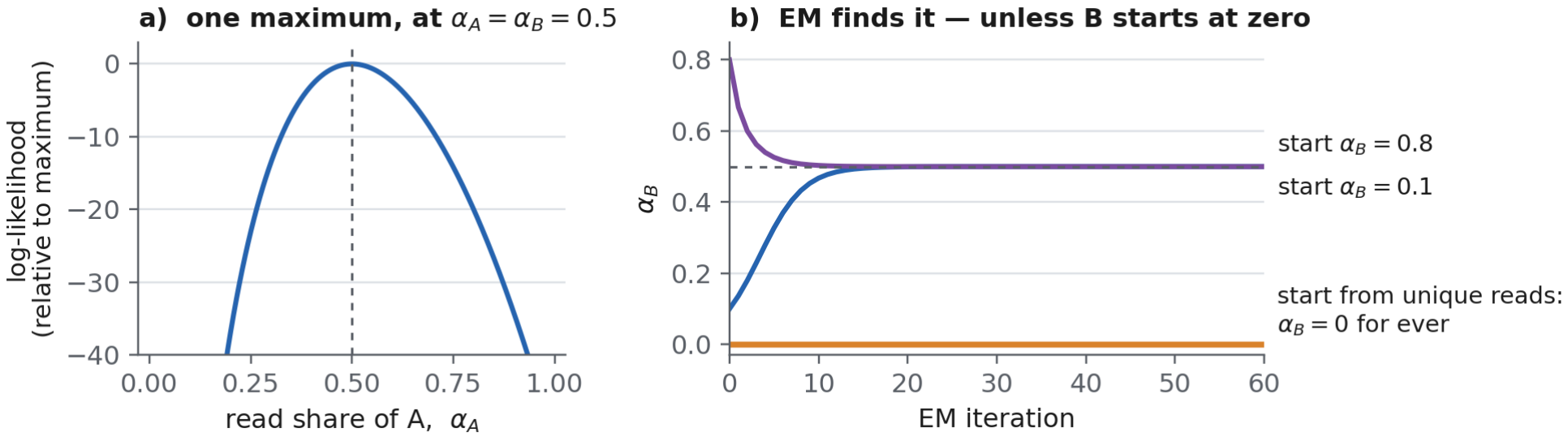
Exercise: A Transcript Without Unique Reads



Transcript B is the first half of A (effective lengths 1000 and 500). 100 reads lie where only A fits, 300 where both fit.

1. How many of the 300 shared reads does the maximum-likelihood solution give to B?
2. EM is started from the unique reads only ($\alpha_B = 0$). What does it return for B?
3. Which transcript has the higher TPM, and by how much?

Answer: B Is Estimated From the Coverage Step Alone



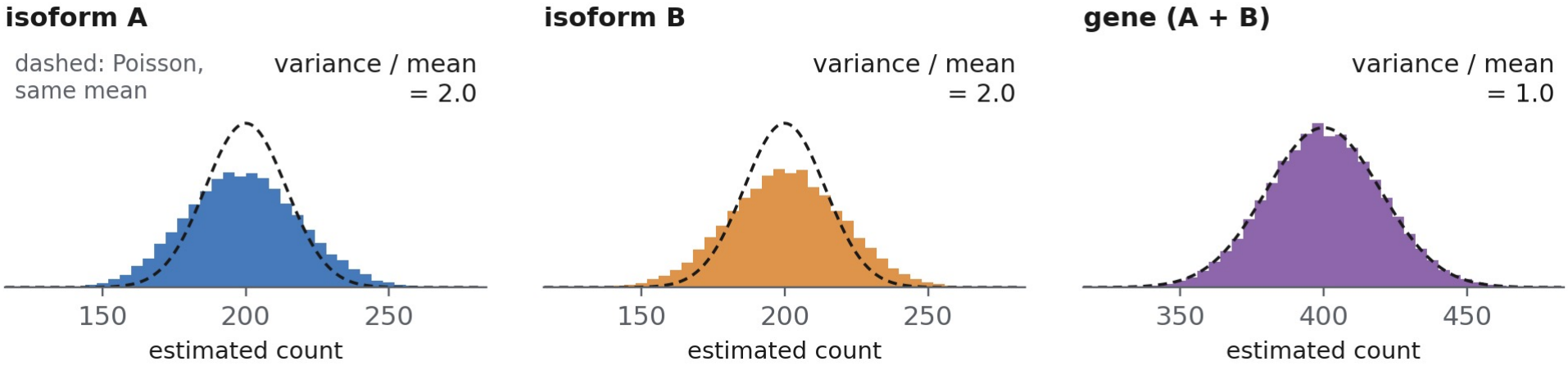
- MLE: B gets 200 of the 300 shared reads, so $\alpha_A = \alpha_B = 0.5$ — and $TPM_B = 2 \times TPM_A$, because B is half as long
- EM converges from any start with $\alpha_B > 0$ (here within ~20 iterations); started from the unique reads only, B stays at 0

→**B rests entirely on the uniform-coverage assumption — like Ccl6-002 (69 %, no unique reads)**

Limitations of Generative Models

- Estimates cannot be correct if the underlying transcript models are incorrect or incomplete
- Abundance estimates are fractions; multiplied by the number of reads they give estimated read counts per transcript or gene. These are not Poisson counts: their error distribution is unclear (next slide)

Estimated Counts Carry Extra Uncertainty

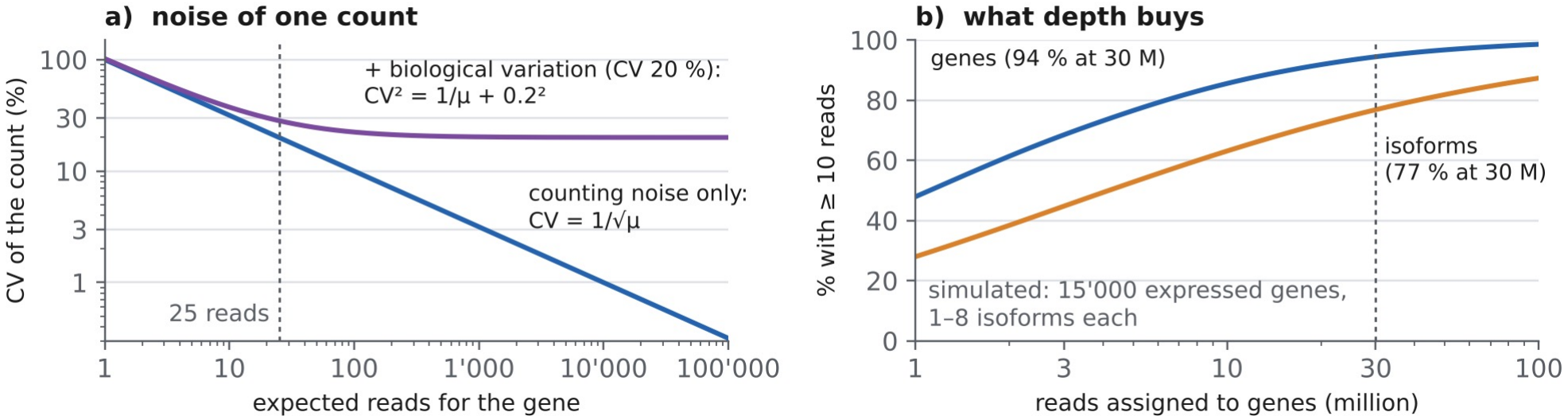


- Bootstrap of the toy example: splitting the shared reads doubles the variance of each isoform (variance / mean = 2 instead of 1); the gene total is unaffected
- The isoform estimates are negatively correlated (here r = −0.5): what one gains, the other loses
- Salmon and kallisto report this as bootstrap or Gibbs samples (inferential replicates)

→ **Use them for transcript-level tests, e.g. edgeR catchSalmon / catchKallisto estimate the overdispersion; divide the counts by it, then analyse as usual**

Baldoni et al. 2024, Nucleic Acids Research 52:e13

Counting Noise: Impact of sequencing depth



- With sampling noise alone, a count with mean μ has CV = 1/√μ: 10 reads 32 %, 100 reads 10 %, 1’000 reads 3 %
- Squared CVs add up: counting noise + quantification uncertainty (previous slide) + biological variation between replicates
- Once biological variation dominates (here above ~25 reads), more depth hardly helps; more replicates do

→ **Isoforms share the gene’s reads and need more depth. ENCODE asks for 30 M aligned reads per replicate**

Implementation of Generative Models

- **RSEM** — Li B, Dewey CN. BMC Bioinformatics 12:323 (2011)
- **Salmon** — Patro R et al. Nature Methods 14:417–419 (2017)
- **kallisto** — Bray NL et al. Nature Biotechnology 34:525–527 (2016)
- **Review** — Pachter L. Models for transcript quantification from RNA-Seq. arXiv:1104.3889 (2011)
- **StringTie** (assembly + quantification of new isoforms) — Pertea M et al. Nature Biotechnology 33:290–295 (2015)
- **Earlier tools:** Cufflinks (Trapnell et al. 2010), MISO (Katz et al. 2010), MMSEQ (Turro et al. 2011), NSMAP (Xia et al. 2011)

Fast approaches to get the Read-Transcript Compatibility Matrix

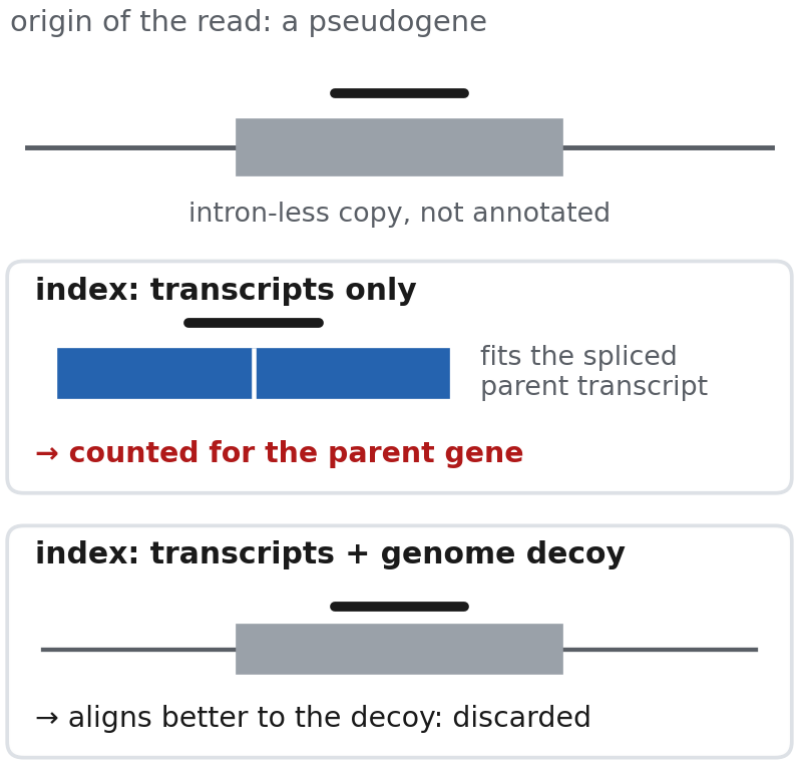
- **Salmon** (≥ 1.0): selective alignment on a pufferfish index
- **kallisto**: pseudoalignment on a coloured de Bruijn graph
- Both return, per read, the set of compatible transcripts (no BAM file)

→**the read–transcript compatibility matrix, condensed into equivalence classes**

Salmon Today: Selective Alignment and Decoys

- Since Salmon 1.0: selective alignment on a pufferfish index (compacted coloured de Bruijn graph); quasi-mapping is retired
- Candidate positions get a real alignment score; mappings below a minimum score are dropped
- Decoys: genome sequence added to the index. Reads that align better to a decoy are discarded

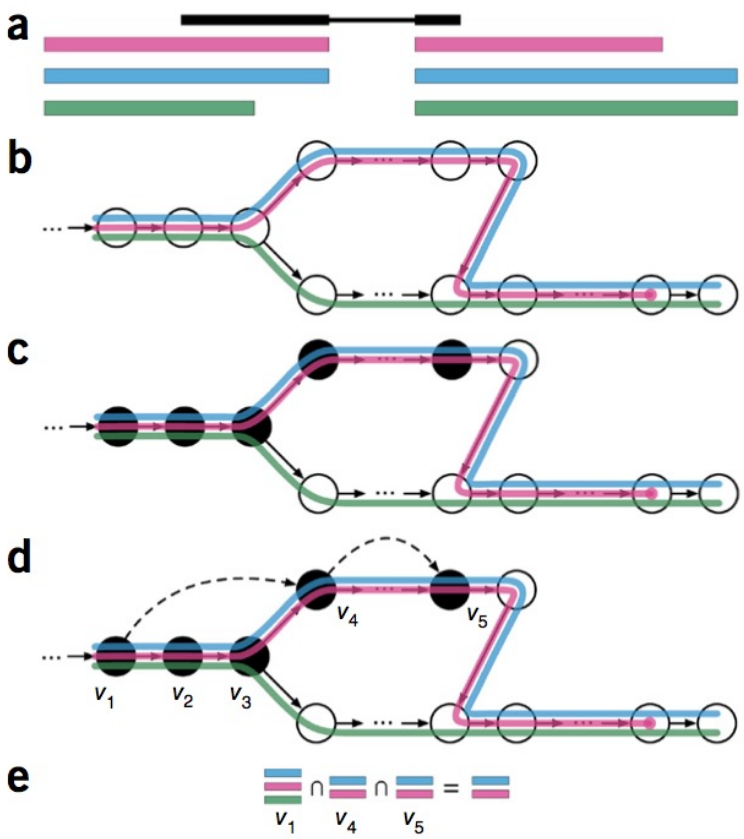
→ **Without decoys, reads from unannotated loci (pseudogenes, introns, intergenic) are forced onto transcripts**



Srivastava et al. 2020, Genome Biology 21:239

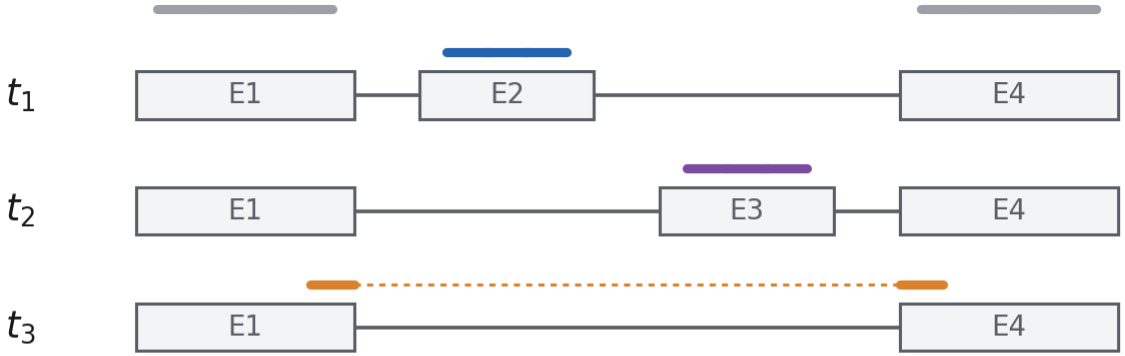
kallisto: Quantification with pseudo-alignments

- Index: a de Bruijn graph of the transcriptome, coloured by transcript
- Find k-mer hits in the de Bruijn graph
- Identifies only transcripts that are consistent with all k-mer hits



Bray et al. Nature Biotechnology, 2016

Equivalence Classes: All the Likelihood Needs

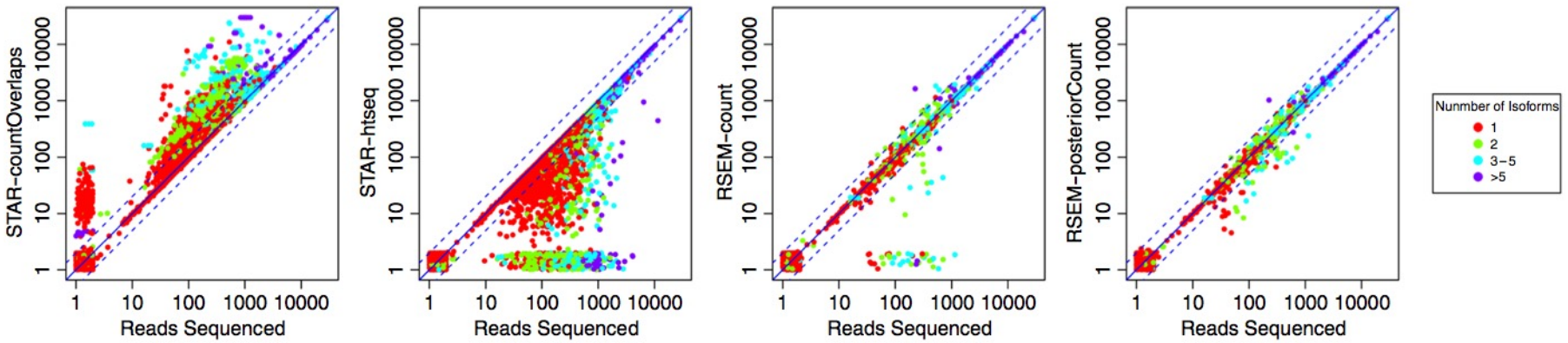


reads are compatible with		count
$\{ t_1 \}$	(in E2)	**120**
$\{ t_2 \}$	(in E3)	**80**
$\{ t_3 \}$	(E1–E4 junction)	**15**
$\{ t_1, t_2, t_3 \}$	(in E1 or E4)	**400**

$$L(\alpha) = \left(\frac{\alpha_1}{\ell_1}\right)^{120}\left(\frac{\alpha_2}{\ell_2}\right)^{80}\left(\frac{\alpha_3}{\ell_3}\right)^{15}\left(\frac{\alpha_1}{\ell_1} + \frac{\alpha_2}{\ell_2} + \frac{\alpha_3}{\ell_3}\right)^{400}$$

- Reads compatible with the same set of transcripts are interchangeable in the likelihood: keep only (set, count)
- EM iterates over classes instead of reads: far fewer, hence fast
- Lost: where in the transcript a read lies. Salmon keeps per-class weights (fragment length, bias): ‘rich’ equivalence classes

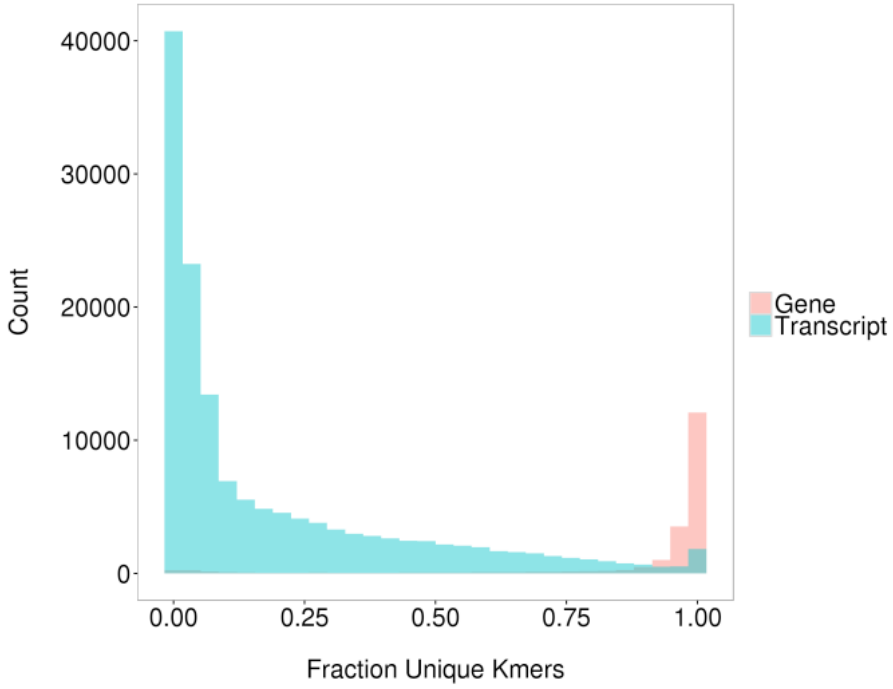
Read Counting Accuracy



- Simulated reads: estimated vs. sequenced reads per gene, for four counting / estimation methods; colour = number of isoforms
- Up to 93 % of the genes are estimated within a factor of 2; the large deviations come from ambiguous alignments
- Isoform differences in at least 109 genes evade detection

Rehrauer et al. 2013, BMC Bioinformatics 14:370 – Blind spots of quantitative RNA-seq

Uniqueness: Isoform-level vs gene-level

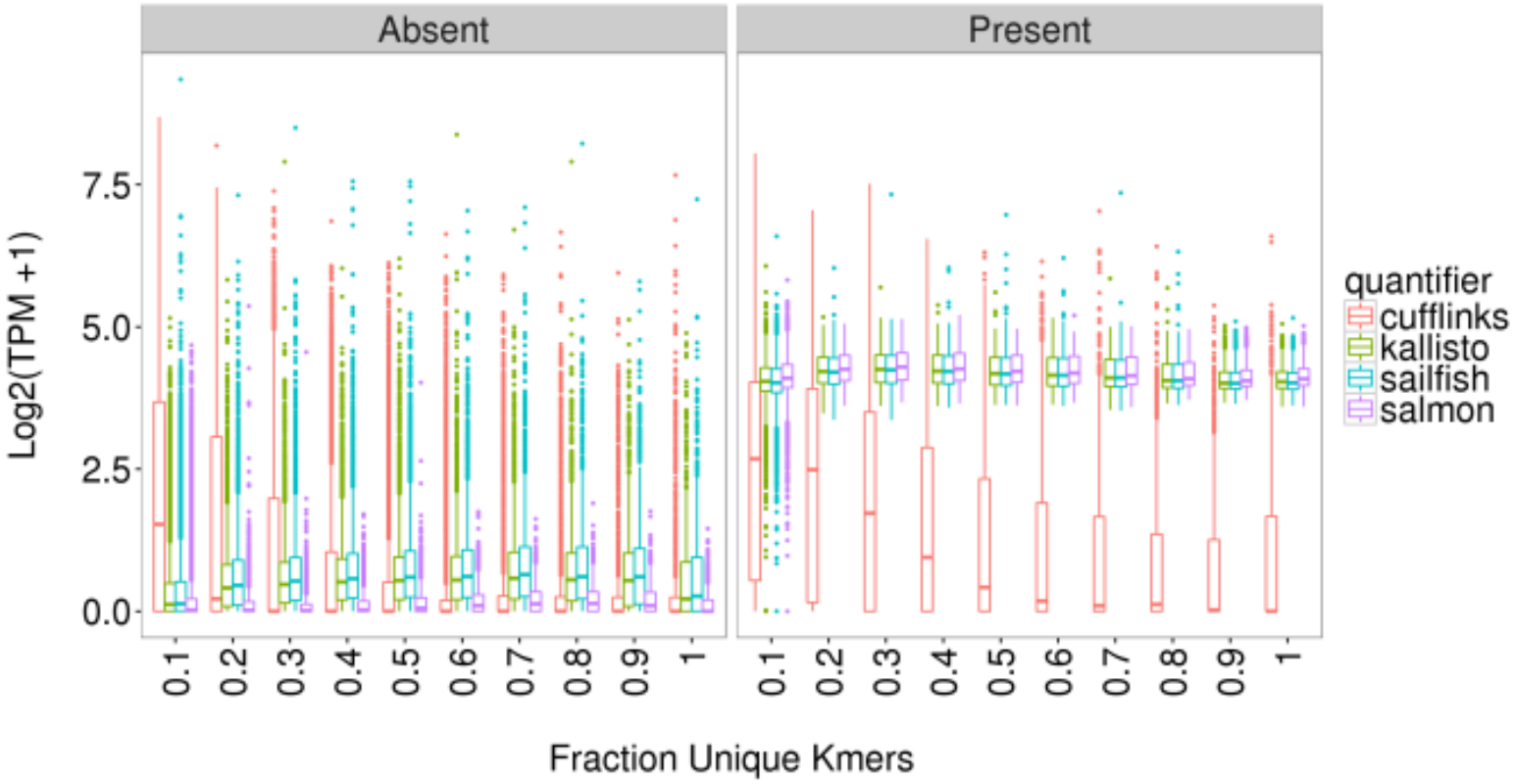


- Fraction of unique k-mer sequences for genes and transcripts
- Ambiguity is mainly between alternative transcripts from the same gene locus

https://cgatoxford.wordpress.com/tag/kallisto/

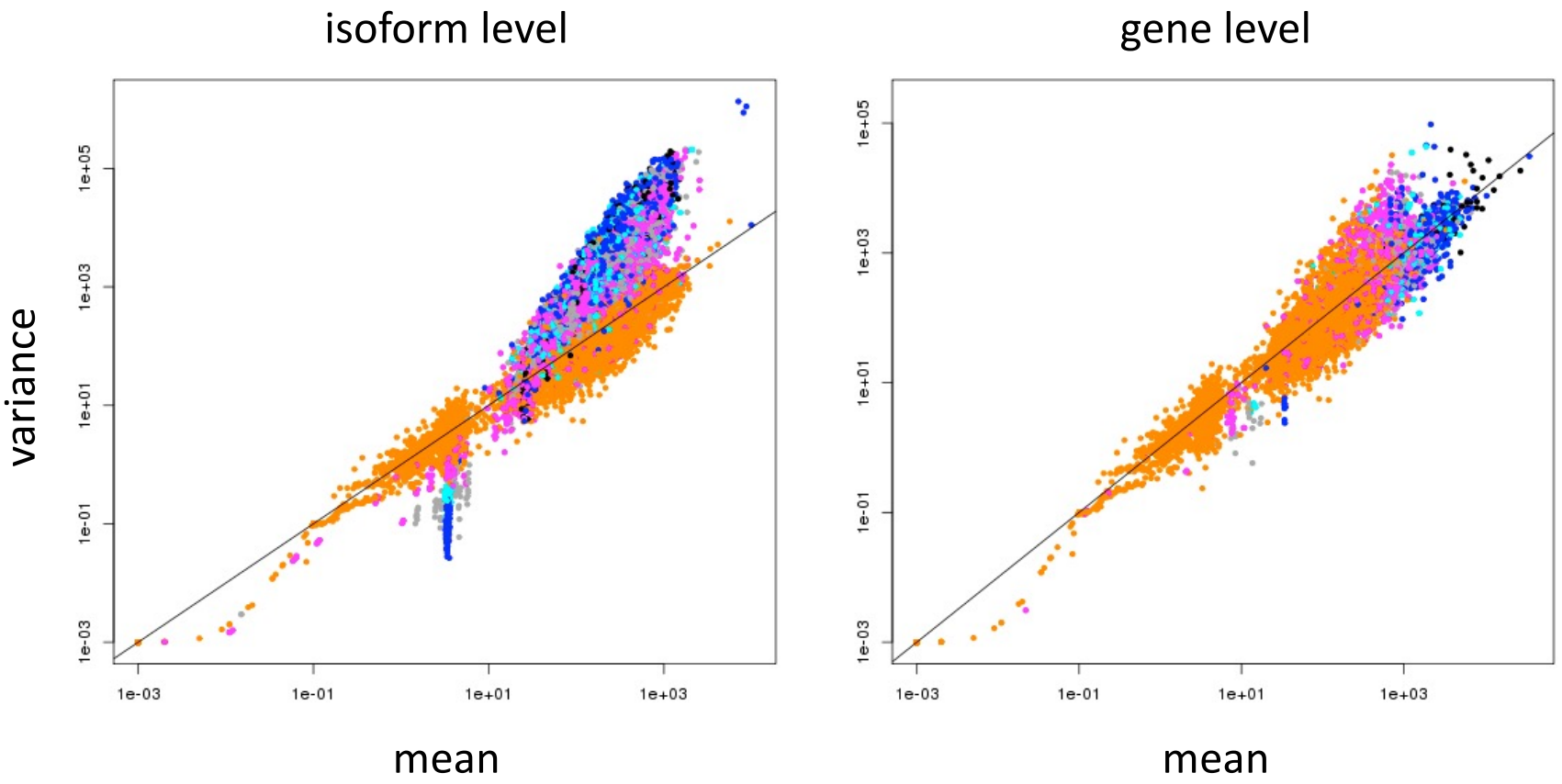
Misassignments to unexpressed transcripts

- Simulated data show that misassignments do happen



https://cgatoxford.wordpress.com/tag/kallisto/

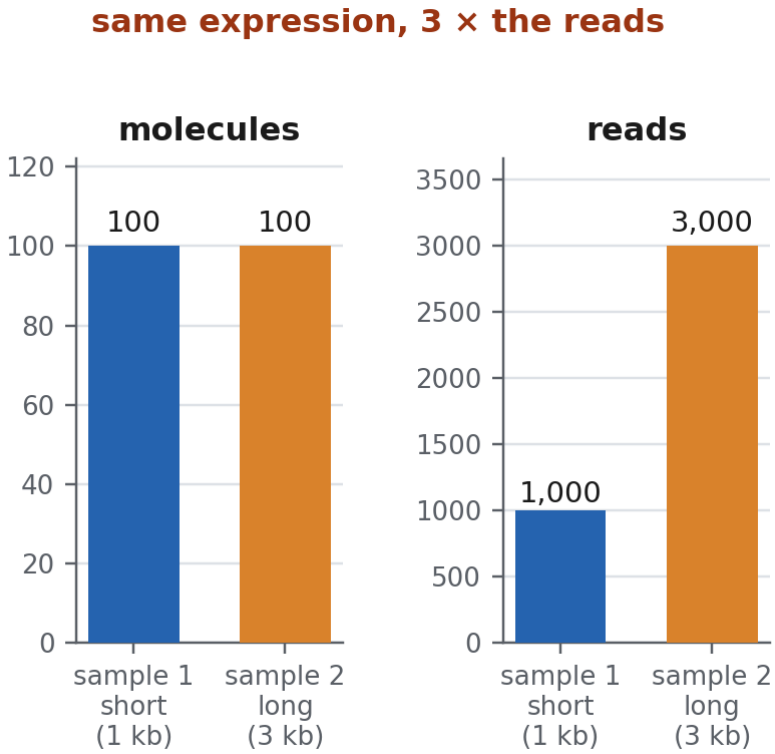
Isoform level has higher variability



Why: isoforms share reads, so their estimates are noisier — the gene sum largely cancels this (cf. the bootstrap example)

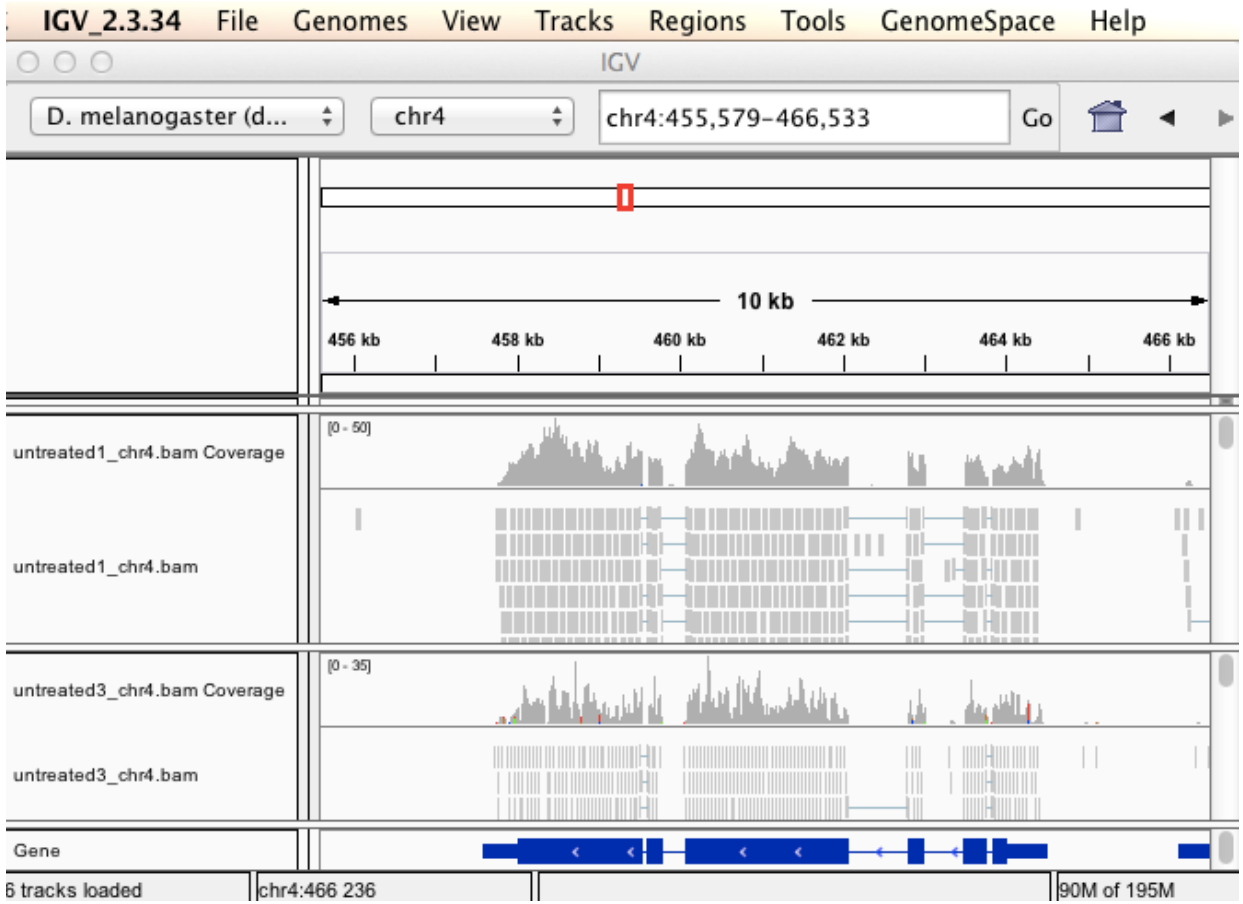
From Transcripts to Genes

- Gene-level estimate = sum over its isoforms: more precise, because their negative covariance cancels
- But reads ≠ molecules: if a gene switches to a longer isoform, its read count rises although its expression does not
- tximport: import estimated counts plus the average transcript length per gene and sample; DESeq2 / edgeR use the lengths as offsets
- Differential transcript usage needs transcript-level methods



Soneson, Love & Robinson 2015, F1000Research 4:1521

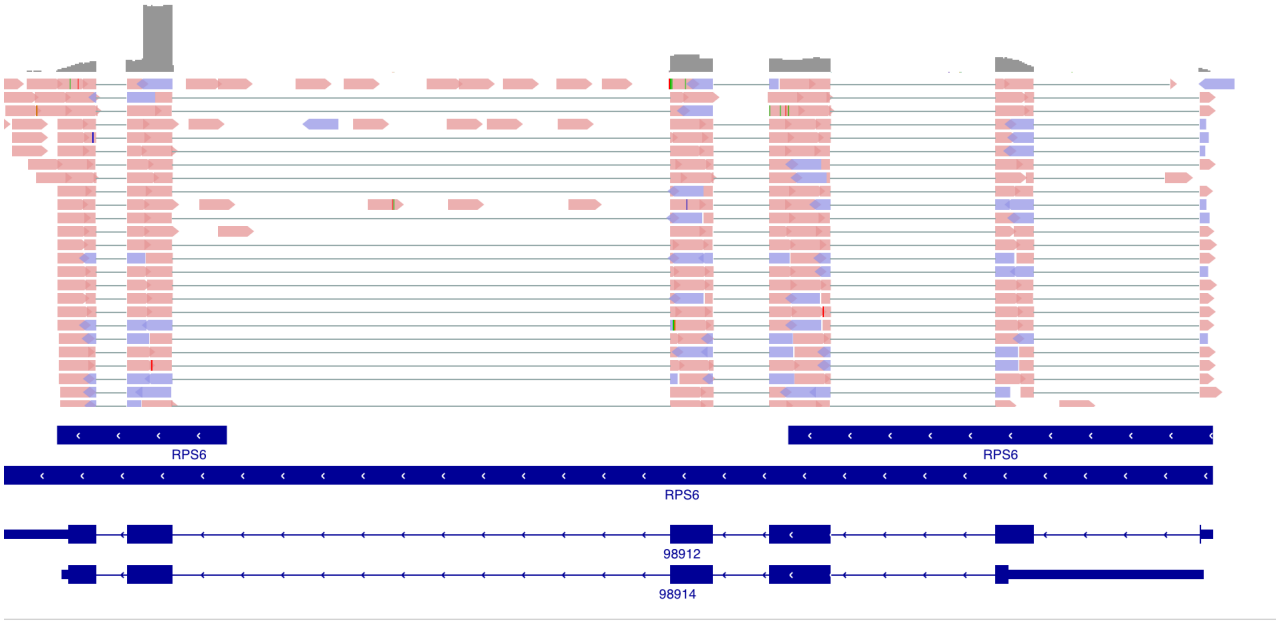
Positional bias of read distributions



- Coverage along a transcript is not uniform
- 3’ bias: degraded RNA, oligo-dT priming
- GC content and random-hexamer priming bias
- **The model splits reads between isoforms assuming uniform coverage — unless the bias is modelled, e.g.** `salmon --seqBias --gcBias`

Highly Duplicated Reads

- Stacks of identical reads (same start, same end)
- Mainly a problem with low input (< 10 ng total RNA): few molecules, many PCR copies
- Highly expressed genes also produce identical reads by chance
- Without UMIs, do not remove duplicates in RNA-seq (lecture 2): it would cap highly expressed genes



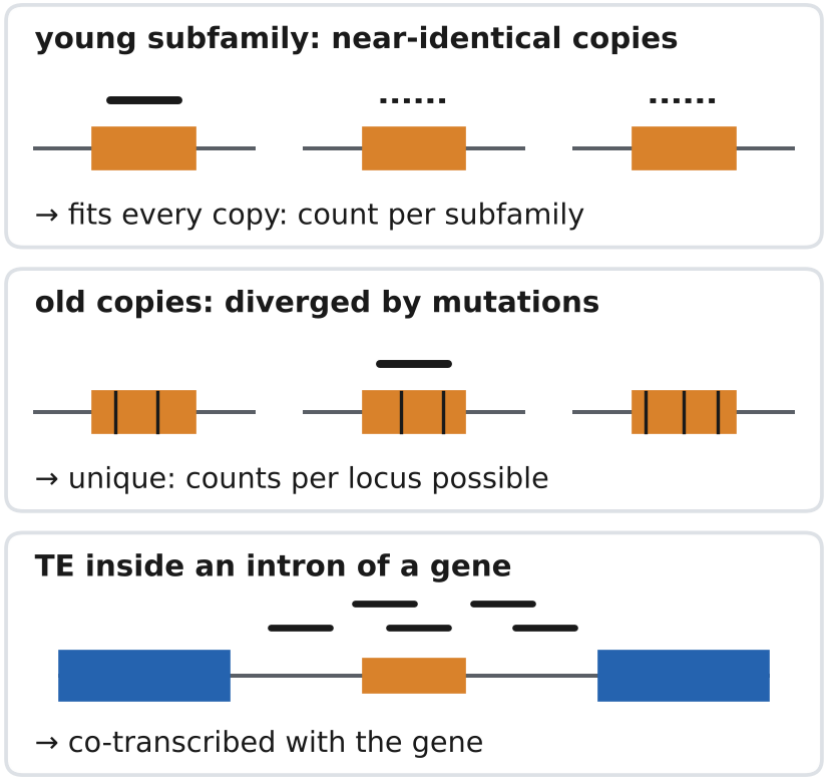
Unspliced transcripts

- Isoform quantification assumes that only mature, spliced transcripts were sequenced
- But unspliced pre-mRNA is captured as well:
  - nascent transcripts in the nucleus, not yet spliced
  - many more with total RNA (rRNA depletion, random priming) than with poly(A) selection
  - poly(A) protocols still pick some up: oligo-dT also primes at A-rich stretches inside introns
- Intronic reads are signal, too: spliced and unspliced counts are the input for RNA velocity (single-cell lectures)

Repeats and Transposable Elements: Multi-mapping Is the Norm

- About 45 % of the human genome derives from transposable elements; copies of young subfamilies (e.g. L1HS, AluY) are almost identical
- Unique-only counting loses many TE reads: allow many alignments (STAR `--outFilterMultimapNmax 100`) and distribute them by EM
- Subfamily level (TEtranscripts) is robust; locus level (TElocal, SQuIRE, Telescope) works only where copies have diverged

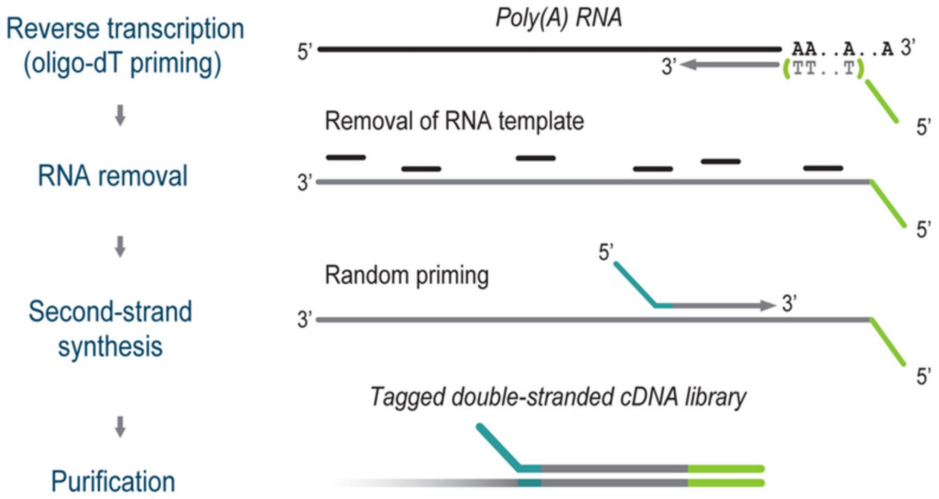
→ **Much of the TE signal comes from TE sequence inside introns and UTRs of genes: compare with the host gene before calling TE activity**



Jin et al. 2015, Bioinformatics (TEtranscripts); Lanciano & Cristofari 2020, Nat Rev Genet

3’-Tagging

- Reads only near the 3’ end: about one read per molecule
- No length normalization: TPM / RPKM do not apply — use the counts
- Isoforms that share the 3’ end cannot be told apart; alternative polyadenylation (different 3’ ends) can
- Reads are assigned to genes



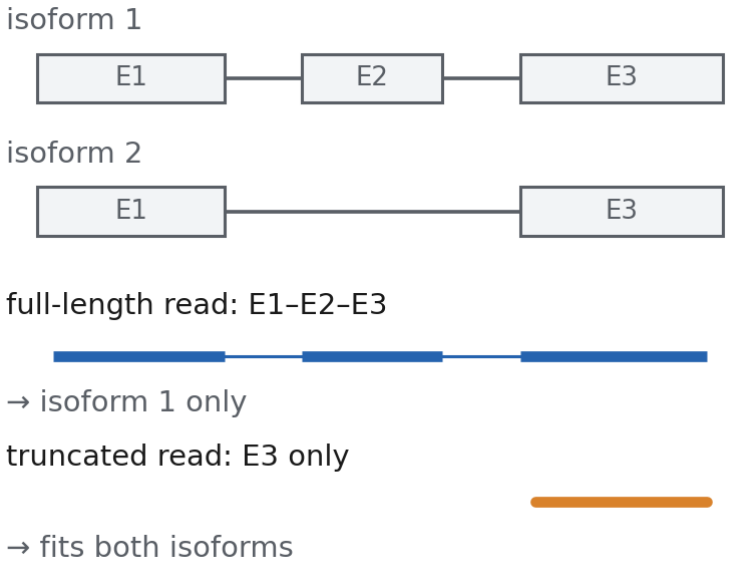
What One Count Means Depends on the Protocol

Protocol	One count is …	Grows with transcript length?	Isoforms
Full-length short reads (e.g. TruSeq)	a fragment from anywhere in the transcript	yes: TPM, length offsets	estimated (EM)
3'-tag (e.g. QuantSeq, BRB-seq)	a fragment near the 3' end	no	only different 3' ends
Droplet scRNA-seq (10x 3')	a molecule (UMI)	no	mostly not
Long reads (ONT, PacBio)	a molecule, ideally full length	roughly no	observed, if the read is complete

- With UMIs, PCR copies collapse to one count per molecule (lecture 2)
- STARsolo, alevin-fry and kallisto | bustools count spliced and unspliced molecules separately: the input for RNA velocity

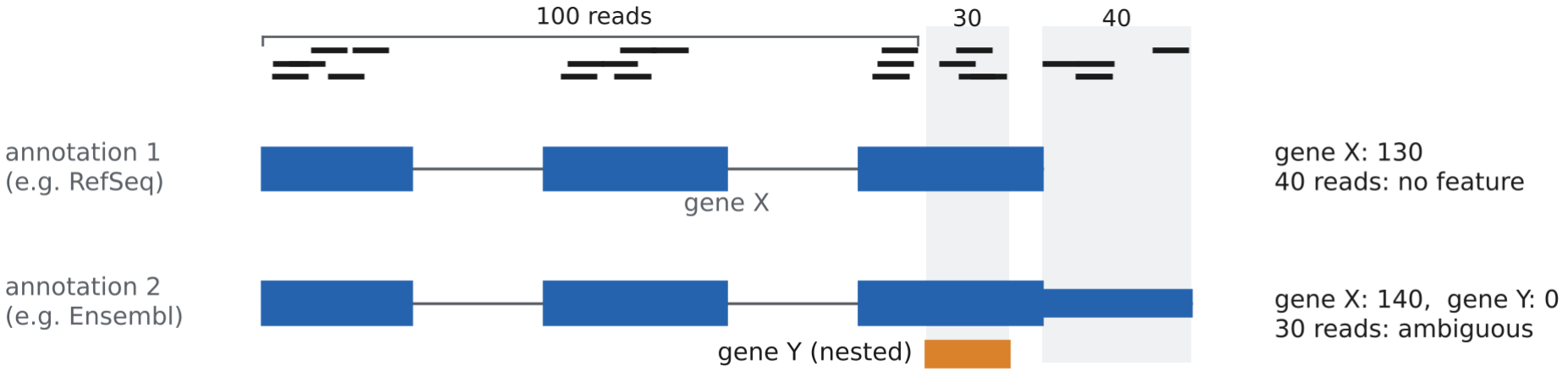
Long Reads: One Read, One Molecule?

- A full-length read shows the whole exon chain: the isoform is observed, not inferred
- Truncated reads (degraded RNA, incomplete cDNA) fit several isoforms: EM is still needed (e.g. bambu, oarfish)
- Lower depth than short reads: fewer counts per gene, noisier quantification
- LRGASP: longer, more accurate reads gave better transcript models than more depth; more depth gave better quantification



Pardo-Palacios et al. 2024, Nature Methods (LRGASP)

The Annotation Is Part of the Model



- Same reads, RefSeq vs. Ensembl: 28 % of genes differ by ≥ 5 %, 9 % by ≥ 50 %
- Why: different gene borders, UTRs and exons, and nested genes that make reads ambiguous (LUZP6 lies inside MTPN: 0 reads with Ensembl)
- Gene counting: a compact annotation (e.g. RefSeq curated) gives more robust counts; isoform quantification needs the comprehensive set (GENCODE/Ensembl), with more ambiguity

→ **Record genome build, annotation source and version; tximeta identifies the transcriptome from salmon's index checksum**

Zhao & Zhang 2015, BMC Genomics 16:97; Love et al. 2020, PLoS Comput Biol (tximeta)

Before Trusting the Counts: QC of the Quantification

Check	Where to look	Warning sign
Mapping rate	STAR Log.final.out; salmon meta_info.json (`percent_mapped`)	low: contamination, wrong species or index, adapters, rRNA
Assignment to genes	featureCounts .summary: Assigned vs. `Unassigned_NoFeatures`, `_Ambiguity`, `_MultiMapping`	many NoFeatures: wrong strand setting, intronic reads (total RNA), genomic DNA
Strandedness	RSeQC infer_experiment.py; salmon in mapping mode: lib_format_counts.json (`strand_mapping_bias`)	≈ 0.5 although the kit is stranded
RNA integrity	RSeQC geneBody_coverage.py, tin.py	coverage piles up at the 3’ end, low TIN: degraded RNA, biased isoform estimates
Composition	rRNA and mitochondrial fractions; share of the top genes	high: failed depletion, stressed or dying cells; fewer informative reads
Fragment length	salmon flenDist.txt; Picard CollectInsertSizeMetrics	very short fragments: degradation; changes effective lengths

→**Compare these numbers across samples: a QC outlier is often the outlier in the PCA**

RSeQC: Wang et al. 2012, Bioinformatics; TIN: Wang et al. 2016, BMC Bioinformatics

Typical setting for bulk RNA-seq

Given a tissue or cell line:

- ~19’400 protein-coding genes (~78’700 incl. non-coding; GENCODE 49)
- ~10 – 18’000 genes expressed at any given time; expression level of the different genes differs largely

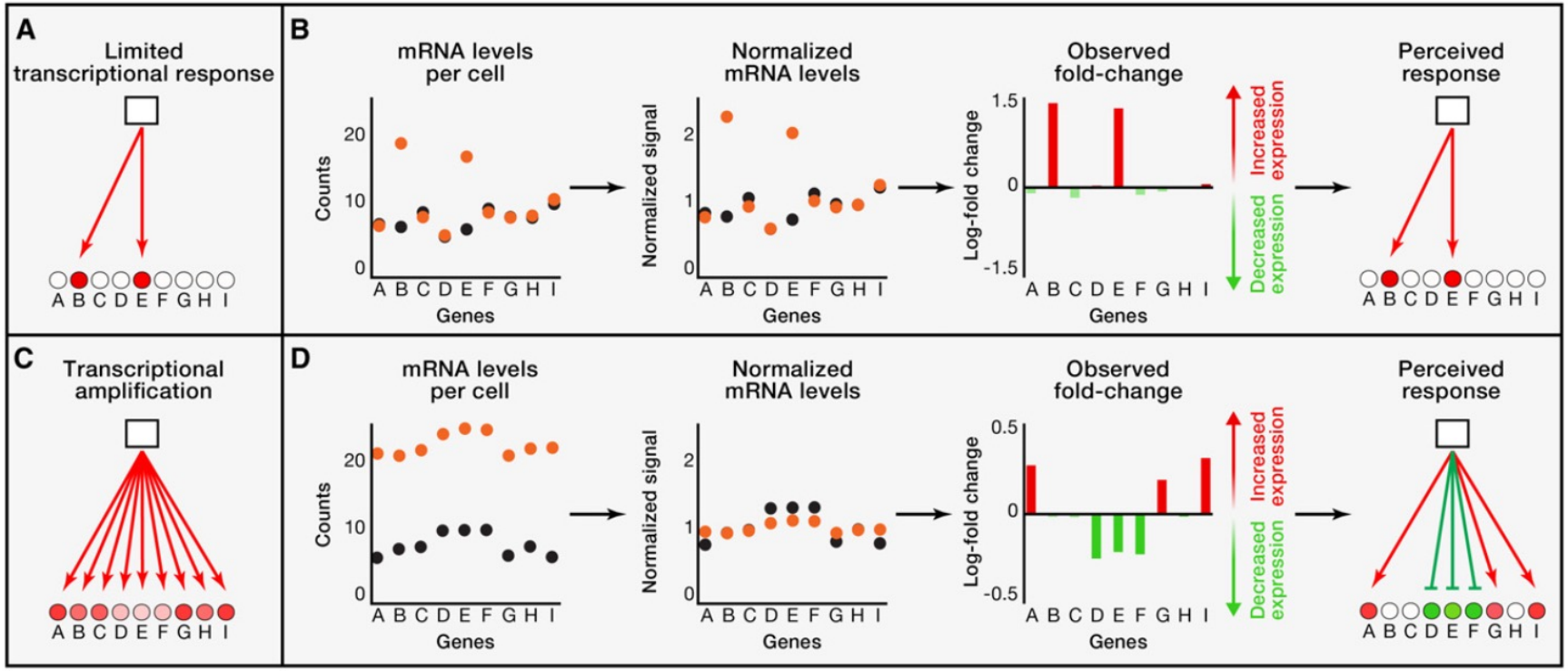
Working hypotheses:

- relative prevalence of a gene reflects activity of associated pathways; more prevalent, more active
- baseline prevalence is not known → **perturbation experiments**: treatment vs control
- the majority of the genes is not affected by the perturbation
- informative quantity: relative abundance of the genes in a tissue
- gene expression has a large dynamic range

Limitations of RNA-seq experiments

- genes vs protein
- alternative explanations …. cell size effects; other side effects
- tissue composition changes

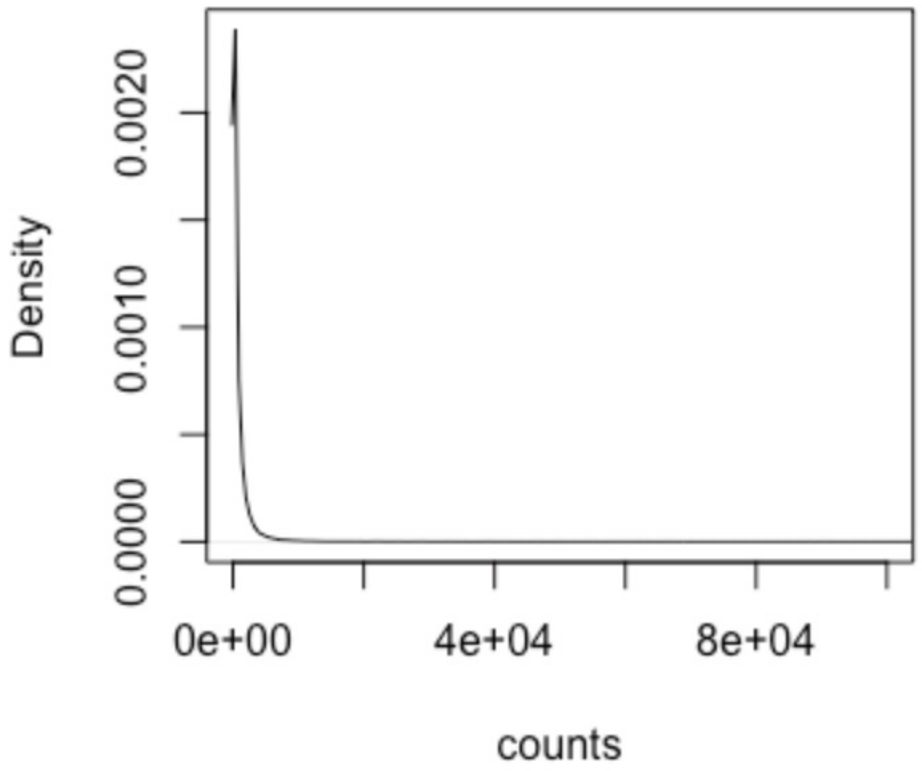
Perturbation Experiments

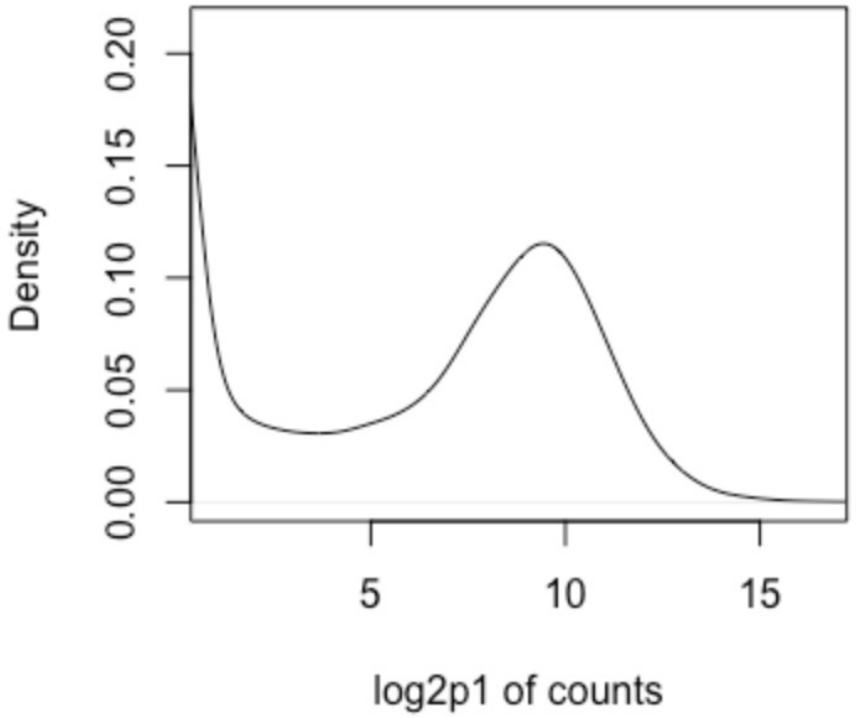


- The top row shows the assumption how a baseline expression profile (black) changes as a response (red) to a perturbation
- Note: expression counts are all around ~10, in practice they range from 0 – 1 Mio

Lovén et al, 2012, https://doi.org/10.1016/j.cell.2012.10.012

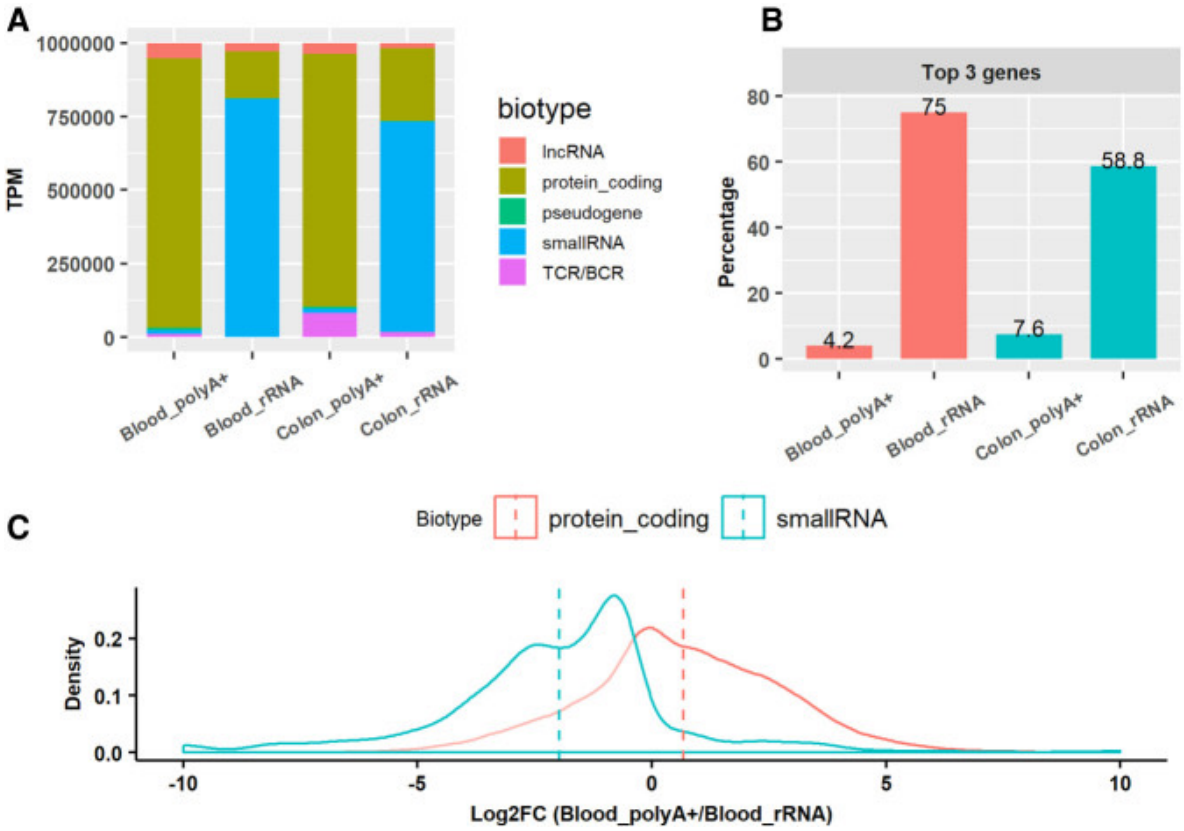
Dynamic Range of Expression and Normalization





TPM normalizes to total count. Is the total count a good measure?

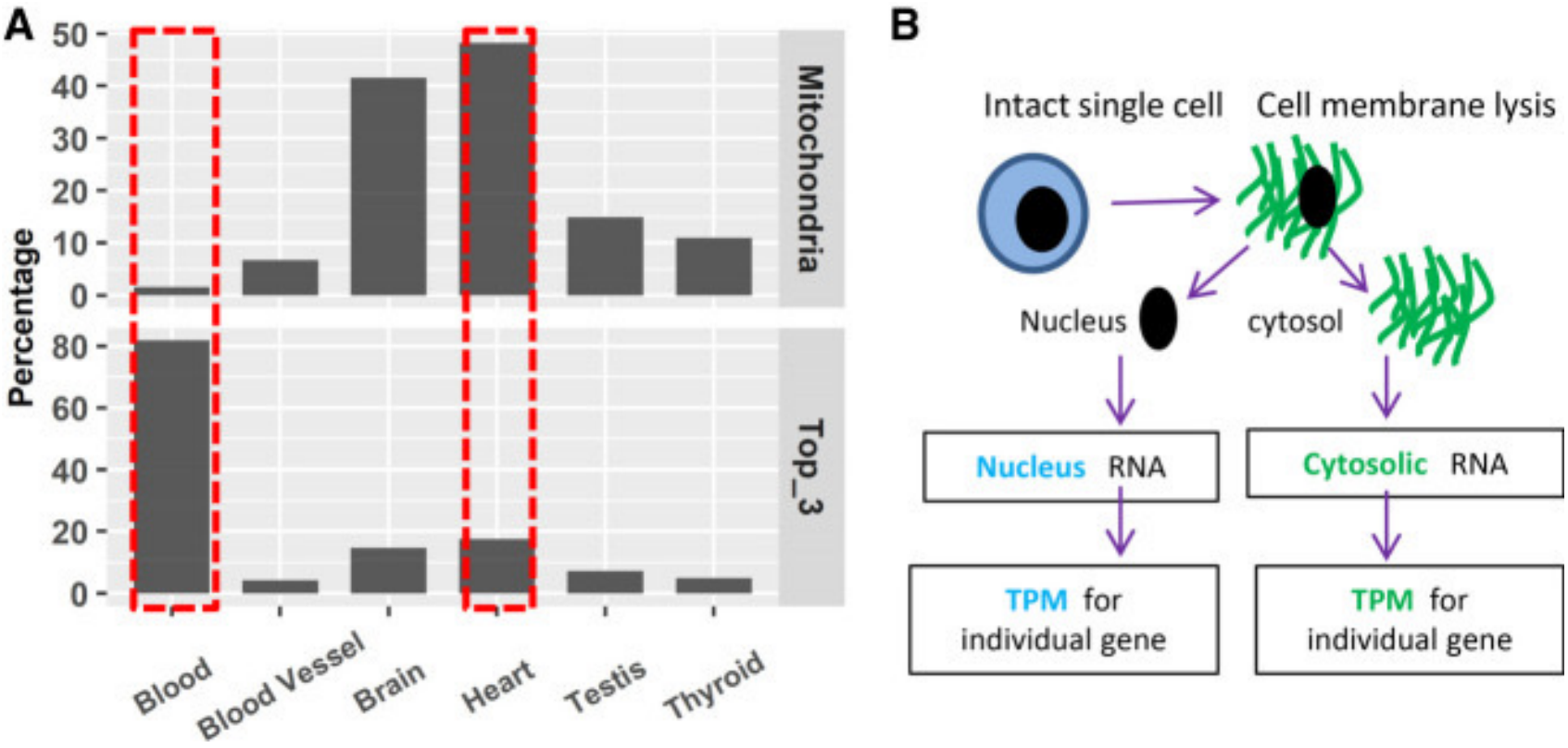
Comparing samples across protocols



- **Issues:** the surveyed RNA populations are not comparable (poly(A)+ vs. rRNA-depleted)
- rRNA-depleted libraries: the top 3 genes take 59–75 % of the total and drive TPM

Zhao, Ye & Stanton 2020, RNA

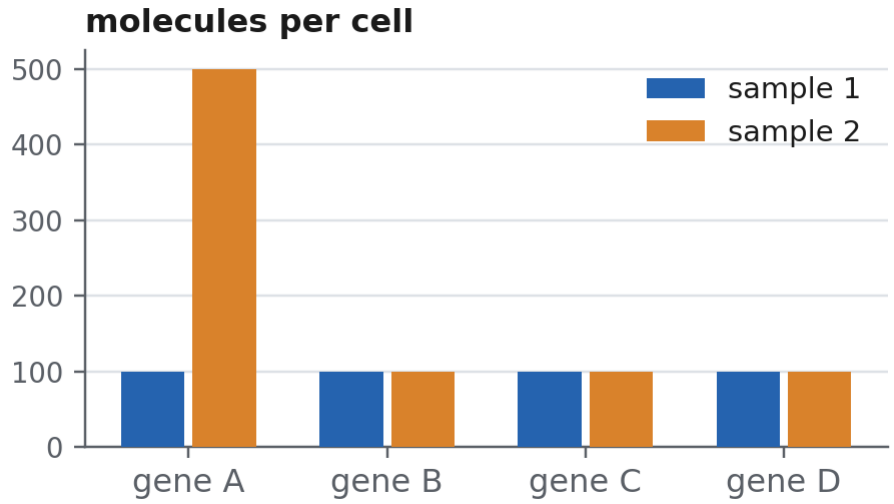
Comparing samples across tissues

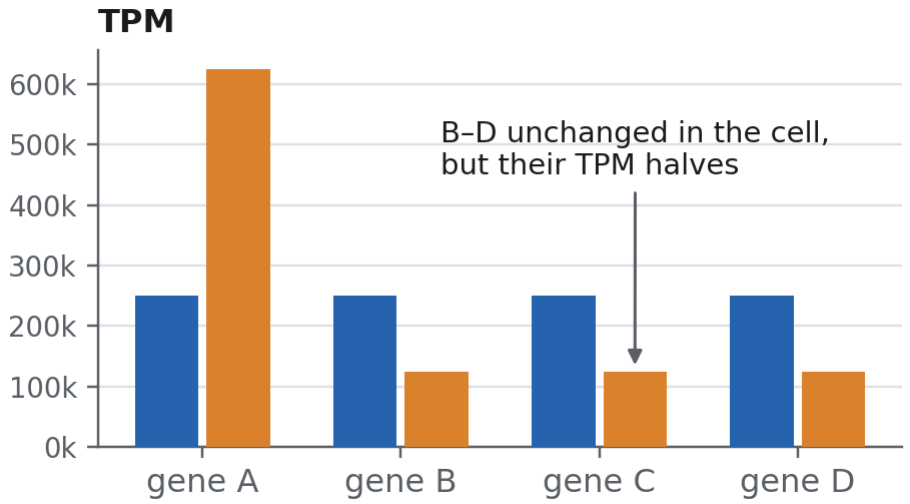


Different tissues may have different populations of genes expressed

Zhao, Ye & Stanton 2020, RNA

TPM Is a Composition: One Gene Moves All Others





- TPM values are shares of a fixed total (10^6): if one gene rises, all others must fall
- Here gene A rises 5-fold; B–D are unchanged in the cell, yet their TPM halves
- Equal TPM ≠ equal molecules per cell when the composition differs (Lovén et al. 2012)

→ **Between-sample normalization (TMM, median-of-ratios) instead assumes that most genes do not change**

Summary

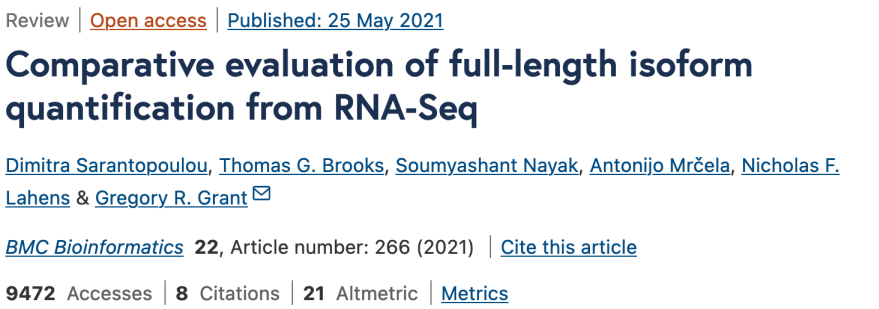
- Gene-level counts are simple, but depend on the counting rules: strand, pairs, multi-mappers
- Generative models + EM split ambiguous reads between isoforms; with alignments you can inspect the reads.
- Estimated counts carry quantification uncertainty; gene-level estimates are more robust than isoform-level ones
- Pseudo- and selective alignment are fast and accurate, but need a complete, correct transcript set (decoys help)
- 3’-tag sequencing gives about one read per molecule: no length normalization, no isoforms beyond different 3’ ends
- Long reads observe isoforms directly, but at lower depth and with truncated reads

References

- Compares kallisto, salmon, featureCounts, … https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-021-04198-1

- from the salmon people, performance evaluation and tuning options (especially genomic “decoy”) https://genomebiology.biomedcentral.com/articles/10.1186/s13059-020-02151-8

- from the salmon people, confirm that decoys help https://www.biorxiv.org/content/10.1101/2021.01.17.426996v1



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Dimitra Sarantopoulou, Thomas G. Brooks, Soumyashant Nayak, Antonijo Mrčela, Nicholas F. Lahens & Gregory R. Grant

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Avi Srivastava, Laraib Malik, Hirak Sarkar, Mohsen Zakeri, Fatemeh Almodaresi, Charlotte Soneson, Michael I. Love, Carl Kingsford & Rob Patro

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Confirmatory Results

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